

NX1005 Specification



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1. Overview

NX1005™ is single chip 2D/3D graphic accelerator including camera controller for handheld information or communication device to implement 2D/3D graphics and camera function easily. It receives commands from external host, reads and writes stacked memory, inputs image data from sensor, and transfers data to main or sub LCD. It's possible to build cheap and high performance graphic acceleration & camera solution with this highly integrated ASIC including 2D/3D graphic engine, texture cache, sensor interface, image signal processor, JPEG compression/decompression engine, PLL, frame buffer, LCD interface and host interface. Flexible sensor interface and various type of host interface can provide customized solution to satisfy any customer's needs including CDMA and GSM.

NX1005™ is designed to support high speed graphic processing, under various aspects of consideration that follows; Firstly, Geometric engine (G-engine) and rendering engine (R-Engine) are operated in pipelining type, based on this, graphic accelerator is around two times faster than the case operated by non-pipelining type. Secondly, there are two operation modes, direct mode and stack mode. Access time of host CPU is reduced dramatically by the stack mode. Thirdly, clipping processing is supported in fully hardwired module, therefore, clipping problem, results in speed-down of 3D games, is automatically resolved. Fourthly, portrait as well as landscape type various 2D/3D contents are be displayed by supporting landscape mode. Finally, NX1005™ supports both graphic functions and camera functions, and hybrid function of graphic and camera engine is also supported by NX1005™.

2. Features

2.1. General Features

- Fully Compliant with OpenGL ES 1.1 and JSR-184
- Full Hardware Pipelined Geometric/Rendering engine
- **5M polygons/s** Triangle Setup Rate (15M vertices/s geometry, 80M pixels/s fill rate@80MHz)
- High Speed Hardware Bone(Skinning, Matrix Palette) Animation
- Hardware Morphing Animation : JSR-184 functions
- Hardware Point Sprite Animation
- High Speed & Memory Saving Buffer Objects(Macro) with Stacked SDRAM Memory
- Multi-Texturing
- Supports 7 Primitives

- Projection & Model View Transform
- View Port Transform
- Supports Lighting up to 8 Light Sources
- Flat/Gouraud Shading
- Culling
- Hardware Clipping Check & Processing
- Texture Mapping
 - Min. Filtering : mipmap, pseudo-mipmap
 - Mag. Filtering : Bilinear
- Texture Combining with Environment Colors
- Fog Processing
- Blending
- Supports various test function (Depth Test, Stencil Test, Alpha Test, Scissor Test)
- Logic Operations
- Supports SVG
- Rich Primitive Drawings
- Lines, Triangles, Circles, etc.
- BitBLT with Transparent Color, Scaling, Scrolling and Rotation
- Supports 3 ROP Operations
- Supports Sprite Engine Image controller for Compact size camera for mobile phone and PDA
- Supports up to 3M pixel sensor (CMOS image sensor)
- Still capture and motion capture application
- Integrated Image signal processor
- Integrated high performance real time JPEG codec
- Direct Main/Sub-LCD preview support up to the half of Sensor resolution
- Various image effects are supported
- 8/9/16/18 bit Parallel and Serial bus support (Intel/Motorola type)
- UART support (Up to 1.5Mbps)
- Internal 2K RAM for firmware (Firmware can be downloaded through parallel bus)
- Maximum flexibility using embedded 8 bit MCU
- Supports Double Buffering
- Up to 320 x 240 Display
- Display Rotation
- Anti-aliasing Filtering
- Supports Hardware Cursor
- Conventional LCD Interface
- Supports Landscape contents
- Supports Graphic and Camera co-working function

- 8 General Purpose Input/Output pins
- Same IO operating voltages can be used for sensor, host and LCD interface
- 0.18um CMOS, 6 layer metal process
- 160 pin CSP uBGA package (9.5mm x 9.5mm)
- Highly integrated low cost solution
- Low power consumption (25mA peak) → TBD
- Power save mode supports: all/sub-block power down (under 100uW) → TBD
- Power down mode supports: PSRAM & PLL Power down & all power down mode (under 50uW) → TBD
- On chip PLL (Input range = 1MHz~30MHz, output range = 50MHz ~ 150MHz)
- On chip Regulator for internal PLL and core voltage (input voltage = 2.8V, output voltage = 1.8V, Output Current = 500mA)

2.2. Detail Features

- 3D Graphics
 - Fully Compliant with OpenGL ES1.1 and JSR-184
 - Supports direct mode and stack mode
 - Full Hardwired Geometric / Rendering engine
 - High Speed Hardware Bone(Skinning, Matrix Palette) Animation
 - Hardware Morphing Animation : JSR-184 functions
 - Hardware Point Sprite Animation
 - High Speed & Memory Saving Buffer Objects(Macro) with Stacked SDRAM Memory
 - Multi-Texturing
 - Supports 7 Primitives
 - Points, Lines, Lines with strip, Lines with loop, Triangles, Triangles with fan, Triangles with strip
 - Supports Lighting up to 8 Light Sources
 - Flat/Gouraud Shading
 - Supports Point/spot/directed light sources
 - Projection & Model View Transform engine
 - Perspective division
 - View Port Transform
 - Culling
 - Hardware Clipping Checking and High speed Clipping processing
 - Texture mapping
 - Min. Filtering : mipmap, pseudo-mipmap
 - Mag. Filtering : Bilinear

Wrapping (Clamping, Repeat)

- Texture data format

8 bits Alpha/8 bits Luminance/Luminance-Alpha 88/RGB 565/RGBA 5551/RGBA 4444

- Texture Combining

Replace / modulate / decal / blend / add

- Texture copy

- Fog processing

- Blending (supports also blending with fixed alpha value)

- Depth test with 8 modes (up to 16 bits)

- Stencil test with 8 modes (up to 4 bits)

- Alpha test with 8 modes

- Scissor test

- Transparent color

- Mask function (Stencil mask, depth mask, color mask)

- Logic operations with 16 modes

- Supports Landscape contents

- Supports co-working of 3D graphic and camera function

■ 2D Graphics

- Supports SVG

- Supports 7 Primitives

Circles, Lines, Lines with strip, Lines with loop, Triangles, Triangles with fan, Triangles with strip

- BitBLT with Transparent Color, Scaling, and Rotation

- Supports ROP3 operations

- Supports Sprite engine

- Supports Landscape concept

- Effective block data transferring among memories

■ JPEG codec

- Integrated JPEG codec

- Integrated JPEG frame buffer

- Various JPEG image size up to UXGA

- Independent JPEG/preview/thumbnail size

- Real time encoding and decoding.

- Support various play speed(x1, x2, x4, x1/2, x1/4) control for MJPEG decoding.

- Up to 30fps MJPEG decoding rate (QVGA)

- Encoder format: YUV422, YUV420

- Decoder format: YUV444, YUV422, YUV420, YUV411.
- User quality control using Q-Table
- Multi capture and motion capture support
- JPEG capture with decoration and special effect

■ Image effect & Image editing

- Up to x4 real zoom (linear steps for vertical and horizontal)
- Color keying with background image or sensor input
- Alpha blending & overlay with background image
- Background image can be apply to any size input resolution
- Background image uses RGB format
- Sketch filter, emboss filter
- Binary, gray, negative image
- Existing RGB/JPEG image can be edited
- All image effects can be applied on partial rectangle area
- Any output format can be generated (RGB, JPEG)
- Input image size allowed up to 2 Mega pixels
- Thumbnail buffer for RGB/YUV or JPEG image
- Programmable thumbnail size.

■ Display controller

- Supports Double buffering
- Hardware cursor (up to 32x32)
- Display rotation
- Anti-aliasing filtering
- Maximum resolution: 320x240

■ Host interface

- Direct support MSM chip set
- Supports parallel (Intel and Motorola style) and UART bus interface
- 16 bits/ 18bits bus interface

■ Sensor interface

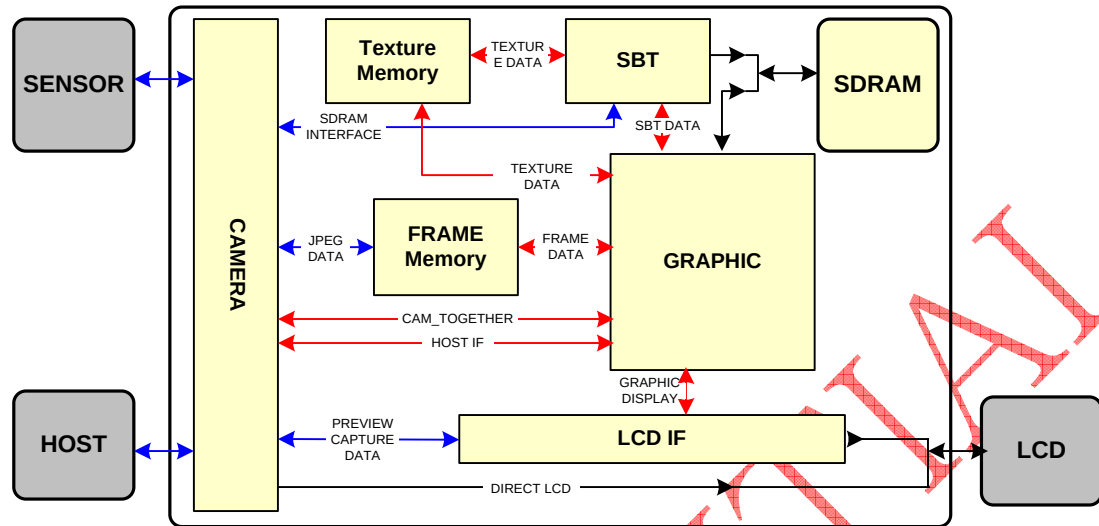
- Supports up to 3Mega pixel (2048 x 1536 resolution)
- Supports CMOS, CCD image sensor.
- Up to 3Mega pixel 30fps operation (Sensor resolution)
- Programmable sensor clock generation
- Various sensor input supports (8 bit YUV422)

- HSYNC, VSYNC, PCLK timing interface
- Programmable sensor signal polarity
- Programmable input data order
- I2C bus master/slaver support
- MCU based flexible sensor control using firmware programming
- Various serial control bus support through GPIO
- Sensor power down control
- Color space conversion
- Test image pattern generation

■ LCD interface

- Self LCD control without host CPU burden
- CPU can access LCD directly through bypass mode
- 8/9/16/18 bit LCD interface for direct preview display
- Main and sub LCD support
- Up to 640x480 resolution true color display
- Various LCD supports (65K, 260K, 16.7M color LCD)
- Programmable RGB order
- Programmable display size
- Up to 30 fps(VGA) preview supported
- Color dithering support for low color LCD
- Supports landscape display

3. Top Block Diagram



[Figure 3-1] GiPump NX1005 top block diagram

The above Figure 3-1 shows top block diagram of GiPump NX1005 in a view.

4. Pin Description

The pin function is defined in 8 categories; System control, host interface, sensor interface, LCD interface, serial interface, power, general interface, and others. The function of each pin is defined in the following table.

Ball	Name	Pad Type	Description
System Control (6 balls)			
A1	RST	IS	Chip reset.
N1	X_IN	I	Clock input (external crystal or Oscillator)
M1	X_OUT	O	Clock output (external crystal)
B2	TMODE2	IS-PD	Operation mode selection TMODE[2:0]. (The signals are float on the PCB.) 000 : Normal mode. 001 : NANDtree. 010 : BIST. 011 : BIRA. 100 : PLL. 111 : SCAN
B3	TMODE1	IS-PD	
C1	TMODE0	IS-PD	
Host Interface (27balls)			
D5	M_INTR	O	Interrupt output to host.
L2	M_CS	I	Chip select from host.
L3	M_WR	I	Write enable from host
D4	M_RD	I	Read enable from host
H3	M_ADDR4	I	(The signal is not used, and should be pulled down on the PCB.)
H2	M_ADDR3	I	Address from host. M_ADDR[3:0]
H1	M_ADDR2	I	
G5	M_ADDR1	I	
G4	M_ADDR0	I	
N3	M_DATA17	B	(In case of not using this signals, should be pulled down on the PCB.)
M4	M_DATA16	B	
G3	M_DATA15	B	Data from/to host. M_DATA[15:0]
G2	M_DATA14	B	
G1	M_DATA13	B	
F5	M_DATA12	B	
F4	M_DATA11	B	
F3	M_DATA10	B	
F2	M_DATA9	B	
F1	M_DATA8	B	
E3	M_DATA7	B	
E2	M_DATA6	B	
E1	M_DATA5	B	
D3	M_DATA4	B	
D2	M_DATA3	B	

D1	M_DATA2	B	
C3	M_DATA1	B	
C2	M_DATA0	B	
Sensor Interface (14 balls)			
J7	S_RSTN	O	Sensor reset control.
J6	S_PWDN	O	Sensor power down control.
C4	S_MCLK	O	Clock out for sensor.
C5	S_PCLK	I	Sensor Pixel Data Sampling Clock (In case of not using this signal, should be connected to ground on the PCB.)
A4	S_HSYNC	I	Sensor Horizontal synchronization pulse. (In case of not using this signal, should be connected to ground on the PCB.)
B4	S_VSYNC	I	Sensor Vertical synchronization pulse. (In case of not using this signal, should be connected to ground on the PCB.)
C7	S_DIN7	I	Sensor data input. S_DIN[7:0] (In case of not using this signals, should be connected to ground on the PCB.)
B7	S_DIN6	I	
A7	S_DIN5	I	
E6	S_DIN4	I	
D6	S_DIN3	I	
C6	S_DIN2	I	
B6	S_DIN1	I	
A6	S_DIN0	I	
IIC Interface (2 balls)			
B5	S_SDA	B	I2C serial data bus
A5	S_SCL	B	I2C serial clock bus
LCD/TV Interface (32 balls)			
M11	TV_CLK	O	TV-Out Clock output
M10	TV_RST	O	TV-Out Reset output
M12	TV_CSB	O	TV-Out Chip Selection
D11	L_CS_M	O	Main LCD chip select.
D10	L_CS_S	O	Sub-LCD chip select.
E11	L_ADS	O	TV-Out/LCD Address/Data selection.
D13	L_WR	O	TV-Out/LCD write enables.
D12	L_RD	O	TV-Out/LCD read enables.
N13	L_DATA23	B-PD	TV-Out/LCD data [23:0]
L13	L_DATA22	B-PD	
L12	L_DATA21	B-PD	
K13	L_DATA20	B-PD	
J13	L_DATA19	B-PD	

H13	L_DATA18	B-PD	
B11	L_DATA17	B-PD	
A11	L_DATA16	B-PD	
H12	L_DATA15	B-PD	
H11	L_DATA14	B-PD	
H10	L_DATA13	B-PD	
H9	L_DATA12	B-PD	
G13	L_DATA11	B-PD	
G12	L_DATA10	B-PD	
G11	L_DATA9	B-PD	
G10	L_DATA8	B-PD	
G9	L_DATA7	B-PD	
F13	L_DATA6	B-PD	
F12	L_DATA5	B-PD	
F11	L_DATA4	B-PD	
F10	L_DATA3	B-PD	
F9	L_DATA2	B-PD	
E13	L_DATA1	B-PD	
E12	L_DATA0	B-PD	
General Purpose IO (8 balls)			
A9	GPIO7	B-PD	GPIO[7]. OSC clock enable (The signal is float on the PCB.)
E8	GPIO6	B	GPIO[6]. External Sub-LCD Chip selection (In case of not using this signals, should be pulled down on the PCB or signal direction should be controlled toward output)
D8	GPIO5	B	GPIO[5]. External ADS (In case of not using this signals, should be pulled down on the PCB or signal direction should be controlled toward output)
C8	GPIO4	B	GPIO[4] (In case of not using this signals, should be pulled down on the PCB or signal direction should be controlled toward output)
B8	GPIO3	B	GPIO[3] (In case of not using this signals, should be pulled down on the PCB or signal direction should be controlled toward output)
A8	GPIO2	B	GPIO[2] (In case of not using this signals, should be pulled down on the PCB or signal direction should be controlled toward output)
E7	GPIO1	B	GPIO[1]. Pulse Width Modulation 1 (In case of not using this signals, should be pulled down on the PCB or signal direction should be controlled toward output)

D7	GPIO0	B	GPIO[0]. Pulse Width Modulation 0 (In case of not using this signals, should be pulled down on the PCB or signal direction should be controlled toward output)
Serial Interface (4 balls)			
L9	SER_CS	I-PD	N.C (The signal should be tied Low.)
M9	SER_CK	B-PD	N.C (The signal is float on the PCB.)
N9	SER_DO	O	UART TX data (The signal is float on the PCB.)
N8	SER_DI	I	UART RX data (In case of not using this signal, should be pulled up on the PCB.)
Power & Ground (21 balls)			
J5	VDDOPM	P	I/O Power for HOST I/F
J11	VDDOPES0	P	I/O Power for internal stacked memory I/F
N2	VDDOPES1	P	
A2	VDDOPS0	P	I/O Power for LCD, Sensor and System I/F
A12	VDDOPS1	P	
N12	VDDOPS2	P	
E5	VDDI0	P	Core power. 1.8V
J9	VDDI1	P	External 1uF ~ 100 uF capacitor is required to reduce the supply voltage spikes
A3	VSS	G	Ground
B1	VSS	G	
B13	VSS	G	
E4	VSS	G	
E9	VSS	G	
E10	VSS	G	
J4	VSS	G	
J10	VSS	G	
J12	VSS	G	
K10	VSS	G	
M13	VSS	G	
K1	AVDD_PLL	P	PLL Analog Power. 1.8V
M1	AVSS_PLL	G	PLL Analog Ground
Total 160 balls = 114 balls + 46 balls(NC)			

I: Input, O: Output, B: Bi-directional, P: Power, G: Ground

S: SCHMIT trigger, PD: Pull own, PU: Pull Up

5. Functional Block Description

5.1. Power Management

5.1.1. Clock Frequency

- Example of Clock setting

Source Clock	PLL Out	System Clock	Graphic Clock	SDRAM Clock	TV-out Clock	Unit
12~30	48	48	48	48	27	MHz
	54	54	54	54		
	81	40.5	81	81		
	96	48	96	96		
	108	54	54	54		
	135	45	67.5	67.5		
	144	48	72	72		

5.1.2. Main Power down Control

Following table 5-2 shows interface block of power management and power saving part of each case.

Name	Description
TV-Clock Enable	External TV Encoder clock enable
External Memory Power Down	Support Deep power down mode for external SDRAM (under 10uA)
Sensor Power Down	Sensor module power down enable
PLL Power Down	Support PLL Power down mode
OSC Power Down	Oscillator pad control enable: Main clock source enable

5.1.3. Sub Power down Control

Following table 5-3 shows internal block of power management and power saving part of each case.

Name	Description
Camera block clock enable	All camera function block clock control
Internal MCU clock enable	Internal MCU(8051) block clock control
Display clock enable	Display block clock control
SBT/SDRAM Controller clock enable	SBT engine & SDRAM controller block clock control

Camera DMA clock enable	Camera DMA block clock control
JPEG encoder clock enable	JPEG encoder block clock control
JPEG decoder clock enable	JPEG decoder block clock control
Sensor interface clock enable	Sensor interface block clock control
3D Mode	
Graphic block clock enable	Graphic block clock control

Power down mode is enabled in hardware. As main power down, all blocks can enter into power down mode using oscillator pad disable control by GPIO. PLL can be also in the lowest power using status while it can be in power down mode before ‘main power down’.

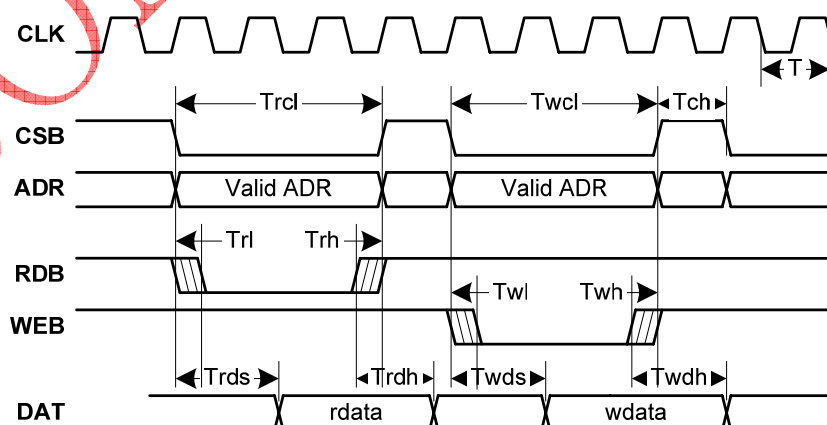
Power down mode can also be entered in software. Camera block and graphic block can be in power down status. Surely system interface and LCD interface is to activate by always-wake clock from the PLL or external oscillator in order to back into “host direct mode” and “system activation”.

Finally gated clock is used for divisional activation of each sub block so that the related function block is controlled to activate.

5.2. Host Interface

NX1005™ supports two kinds of host interfaces, Parallel and UART bus interface. Users can use parallel host interface, Intel type. Host interface is defined by the value of two external pin signals, L_S_0 and L_S_1. When NX1005™ is configured as parallel host interface, user can select 8 bits or 16 bits interface through register programming.

Parallel Bus Interface Timing



[Figure 5-1] Host interface Write/Read Timing

Host interface timing parameter

Parameter	Symbol	min	typ	Max	Unit
Clock	T	8.3	–	78	ns
CSB active period	Twcl / Trcl	2*T	–	–	ns
CSB inactive period	Tch	1*T	–	–	ns
CSB low to WEB/RDB low*	Twl / Trl	0	–	–	ns
CSB high to WEB/RDB high	Twh / Trh	1*T	–	–	ns
Write/Read data setup time	Twds/Trds	0	–	–	ns
Write/Read data hold time	Twdh/Trdh	2	–	–	ns

T: internal clock cycle time defined by PLL setting

[Table5-1] Host interface Write/Read Timing parameter

5.3. Register Programming

Registers of NX1005™ can be set by internal or external MCU. Programming method for each case is as below.

External MCU: The registers in NX1005™ can be programmed with external MCU via multiplexed parallel bus interface.

i) **Graphic Registers** – Graphic related registers consists of direct control registers and indirect control registers. Direct control registers are programmed by transferring register address and data together. And indirect control registers are programmed by transferring indirect register data after setting indirect address.

ii) **Camera Registers** – For the programming of Camera related registers, select bank of register and write one byte register address first and write register value next. To define the bank, three address bit A[3:1] are used. Final 4'th bit A[0] out of four address A[3:0] is used to select between register address and register value. Register address byte should be written during A[0] pin is "high". Register value can be read or written during A[0] is "low". Only image data can be accessed with full 16 bits data bus.

Internal MCU: The individual registers in NX1005™ can be accessed by the internal 8-bit MCU using external address access command. Access method is same and only 8 bits width is allowed even accessing image data.

5.4. Graphic Function

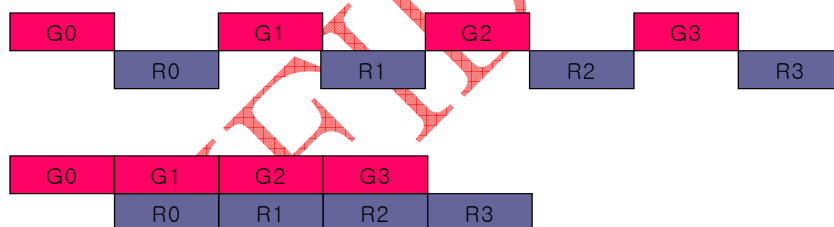
NX1005™ is mainly and initially designed in three aspects; high speed graphic data processing, various contents support and co-work with camera functions. As the results, NX1005™ has strength in these graphic functions.

5.4.1 Operation mode of 3D graphics

NX1005™ supports two types of operation mode. These are direct mode and stack mode. Direct mode means that external MCU transfers data of vertex, color, texture axis and normal values to NX1005™ directly and one by one. The other one is stack mode, which means that external MCU transfers only the start address of index array to NX1005™ and NX1005™ reads data of vertex, color, texture axis and normal values from stack memory. Operation in stack mode enables Read/Write access time of external MCU and produces good result of fast speed.

5.4.2 Pipelined hardware G-engine and R-engine

Geometric engine and rendering engine of NX1005™ are designed as full hardware, operating in pipelining types. As shown in figure 5-2, pipelining mode operation requires about half time of non-pipelining mode operation.



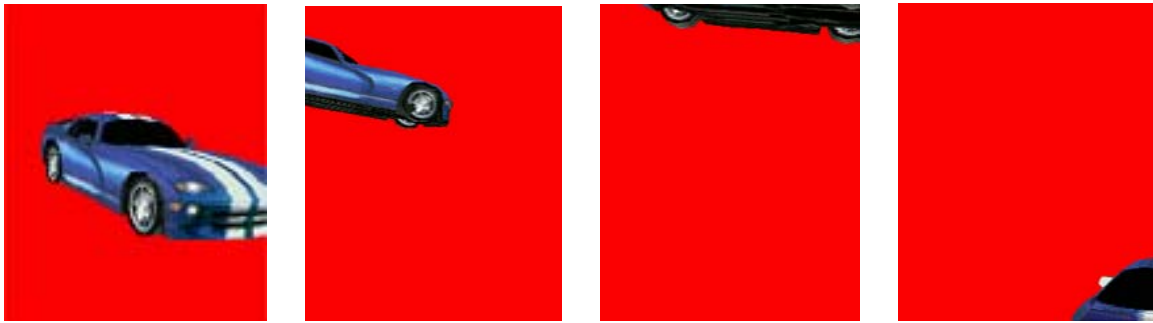
[Figure 5-2] Processing time comparison of non-pipelined and pipelined operation mode between geometric and rendering engine

Where, G1 means geometric processing time of No. 1 polygon and R1, rendering time of polygon processed at G1.

5.4.3 Hardware Clipping

Relevant clipping problems in 3D game are critical cause of speed decrease. Potential clipping matter is resolved in hardware on-chip of NX1005™. It results in graphic processing speed decrease and requires additional burden to external MCU, when clipping checking is done in hardware and clipping processing is done in external MCU.

NX1005™ processes both clipping checking and clipping processing on-chip so that the speed-decrease matter is perfectly resolved in NX1005™.



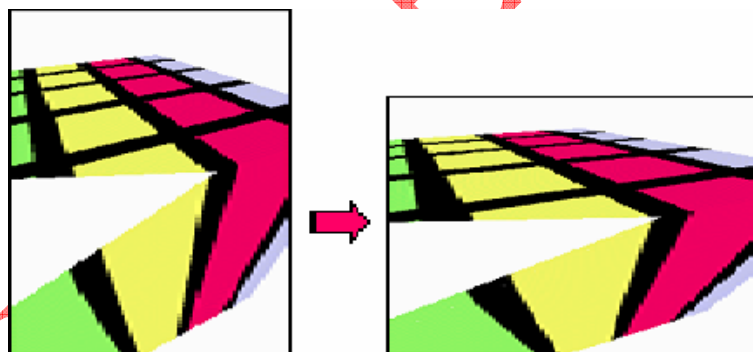
[Figure 5-3] clipping process capture demo

5.4.4 Landscape

NX1005™ is designed to support various contents. Landscape based contents can be used through little setting job. Differ to simple angle rotation, support of landscape requires additional circuit on hardware.

The following figure 5-4 shows portrait mode and landscape mode of a same content. The coordinate ratio of display is same in portrait mode and landscape mode.

Following pictures are display result of one object made on portrait mode (figure 5-5, left) and landscape mode (figure 5-5, right). The starting point of both contents is left right point.



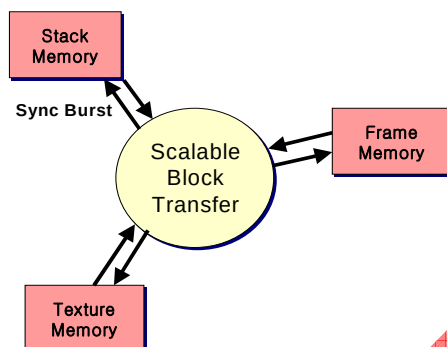
[Figure 5-4] Portrait and landscape display of a same content



[Figure 5-5] portrait and landscape display of a same object

5.4.5 Effective block data transferring among the stack memory, texture cache and frame memory

NX1005™ is on-chip memory includes frame memory and texture cache, and supports effective block data transfer between on-chip memory, or between on-chip memory and stack memory. This can be used for various applications like bind texture, copy texture, Stretched BitBLT, sprites. In terms of block data transfer with stack memory, partial block data transfer can be supported by NxNX1005™, which can't be supported in simple sync burst.



[Figure 5-6] Memory block transfer diagram

5.4.6 Hybrid function between Graphic and Camera

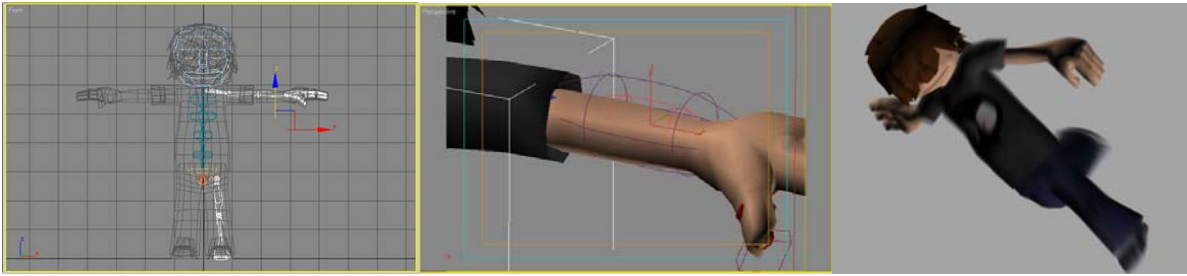
NX1005™ is designed to support both graphic function and camera function. Various applications can be made based on this function. For instance, camera preview image is showed on moving 3D display on LCD panel, instead of simple LCD display. A real like 3D Avatar can be created based on camera capturing feature. 3D game can be played with background preview mode, so that the game users feel like the game objectives are real moving on the very spot. (Patent applied)



[Figure 5-7] 3D graphic content with background preview image

5.4.7 Bone(Skinning, Matrix Palette) Animation

NX1005™ has some special 3D animation functions. One of them is a “Skinning” animation. Skinning is a term used a lot in current computer and console game programming tutorials. It provides a way to approximate how character skin and clothing reacts to the movement of “bones” inside the character. It allows using soft bends in character joints (such as elbows and knees) instead of ugly, blocky, and sharp angles and holes that used to cause problems in older renderer implementations without skinning support. Especially this animation is essential in the case of high quality “mobile” 3D game design.



5.4.8 Morphing Animation

NX1005™'s another 3D animation function is “Morphing”. It defines multiple vertex positions for single set of polygons. The final vertex position can be calculated by interpolating the vertex positions between the default position and the morph position(s) to create facial expressions and other smooth deformations without huge amount of vertex positions.



5.4.9 Point Sprite Animation(effects)

Sprites are useful for speeding up rendering when real 3-D geometry is not really needed (such as very distant objects). They are also useful for special effects, such as lens flares or particle clouds, smoke, or water bubbles.



Particle effect by using Sprites

5.5 Camera Function

5.5.1 Image Sensor Interface

NX1005™ supports one sensor type, no ISP sensor. The sensors with built-in ISP have 8 bits YUV422 output, and the sensors without ISP have 10 bits raw RGB output in normal case. The sensor data interface of NX1005™ is selected through register programming. Operation speed of sensor is also controlled by NX1005™.

The sensor is controlled by internal MCU through I2C interface. Host CPU can access registers of sensor through NX1005™ to minimize CPU burden. NX1005™ can control any sensor that does not use I2C interface through general-purpose pins. NX1005™ has sensor power down pin and clock control logic for sensor power control.

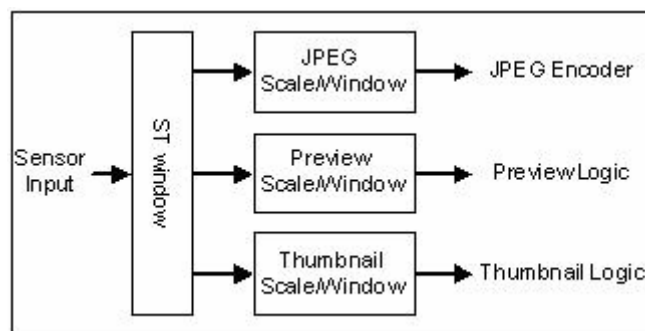
Vendor	Parts Number	Remark
Micron	MT9V111 (VGA) MT9M111 (1.3M)	
OmniVision	OV7648 (VGA) OV9640 (1.3M) OV9650 (1.3M) OV2620 (2.0M)	
IC-Media	ICM-105C (VGA) ICM-105T (VGA)	
Zoran	ZR32212 (1.2M) ZR32316 (2.1M)	
PixelPlus	PO1030K (VGA) PO2130K (1.3M) PO3130N (1.3M) PO1200K (2.0M)	
Hynix	HV7131xx (VGA) HV7161SPA1 (1.3M)	
SAMSUNG	S5K532EA (VGA) S5K3A1EA (1.3M)	
C.I.Sensor	CIS-VV15 (VGA)	

[Table 5-2] sensor list supported by NX1005™

5.5.2 Image Window Size Control & Scaling

There are 4 different window control logics in NX1005™.

- Start window size control from sensor image (Not scaling – can use scaled sensor image).
- JPEG image scale and window.
- Preview image scale and window.
- Thumbnail image scale and window.



[Figure 5-9] Window Control in NX1005™

ST window logic is used first to get internal initial image size of NX1005™ from original image size of sensor. The size will be 2048 x 1536 for 3mega pixel sensor. NX1005™ has three scale-window logics for JPEG, preview and thumbnail. Each image size can be set independently. JPEG size is allowed up to 2048 x 1536, preview size up to 240 x 320, and thumbnail size up to 128x128. See [Figure6-6] below.

The scaling process is applied prior to windowing process. Scale factor can be selected in any value to achieve smooth scale function.

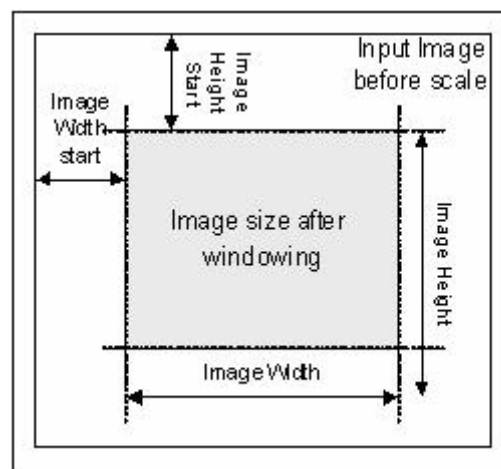
The following table is a zoom control example from 1280 x 1024 sensor input image to 120 x 120 preview image.

Input size	Scale factor	Size after scaling	Window size	Final zoom effect
1280 x 1024	1/8	160 x 128	120 x 120	100%
1280 x 1024	1/7	183 x 145	120 x 120	113%
1280 x 1024	1/6	214 x 171	120 x 120	133%
1280 x 1024	1/5	256 x 205	120 x 120	160%
1280 x 1024	1/4	320 x 240	120 x 120	186%
1280 x 1024	1/3	427 x 342	120 x 120	267%
1280 x 1024	1/2	640 x 480	120 x 120	375%
1280 x 1024	2/3	854 x 683	120 x 120	534%
1280 x 1024	3/4	960 x 768	120 x 120	600%
1280 x 1024	4/5	1024 x 820	120 x 120	640%

1280 x 1024	6/7	1198 x 878	120 x 120	686%
1280 x 1024	9/10	1152 x 922	120 x 120	720%
1280 x 1024	15/16	1200 x 960	120 x 120	750%
1280 x 1024	31/32	1240 x 992	120 x 120	775%
1280 x 1024	1	1280 x 1024	120 x 120	800%

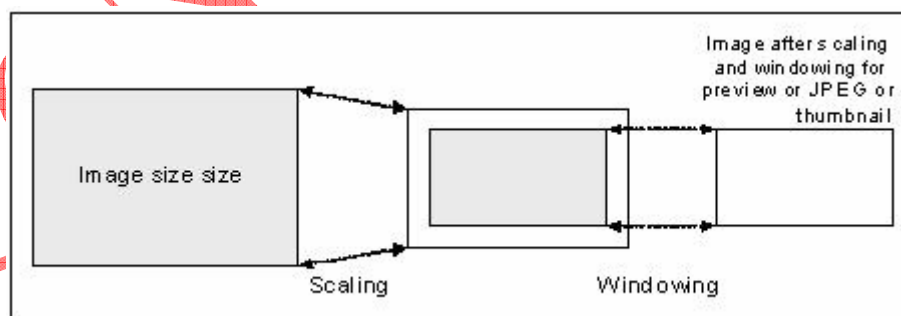
[Table 5-2] Zooming result by scaling factor

Windowing is processing with image width start cut, image height start cut, and image width and image height parameters. User can implement digital zoom or pan function with scale and window programming



[Figure 5-10] NX1005™ Image Window Size Control

With scale and window size control, user can define the proper JPEG, preview or thumbnail image size with various zoom effects. [Figure6-8] shows whole scaling/window process.



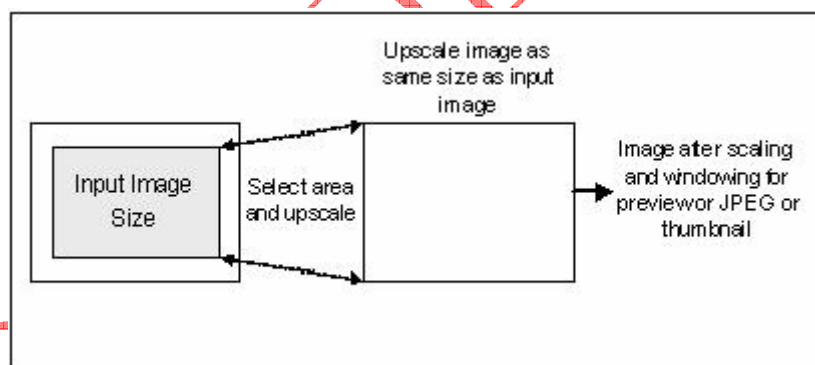
[Figure5-11] Image Scaling & Window definition of NX1005™

5.5.3 Auto-Focusing Control

NX1005™ 3x3-matrix calculation functions are to implement BMP (Band Pass Filter) and edge detection filter. NX1005™ uses only Y values for every pixel to calculate edge values. The summation of BPF and edge value of every sensor frame from this filter is stored in registers. The filter value and two windows for AF summation can be defined. Internal MCU or external host should read these summation values to decide, whether the focus is good or not. If current focus is not good enough, users can control motors by lens moving. The Focusing control uses PWM method and I2C.

5.5.4 Input Image Upscale

NX1005™ has special digital zoom out logic and linear input image upscale. Users can implement digital zoom function described as above “Scaling and Preview zoom control” section. But also, NX1005™ has real digital zoom function with this image upscale logic. User can obtain enlarged picture of same image size as that with sensor input, not just preview size image. NX1005™ supports up to x4 digital zoom with 32 steps. Users can implement digital zoom function like normal digital still camera

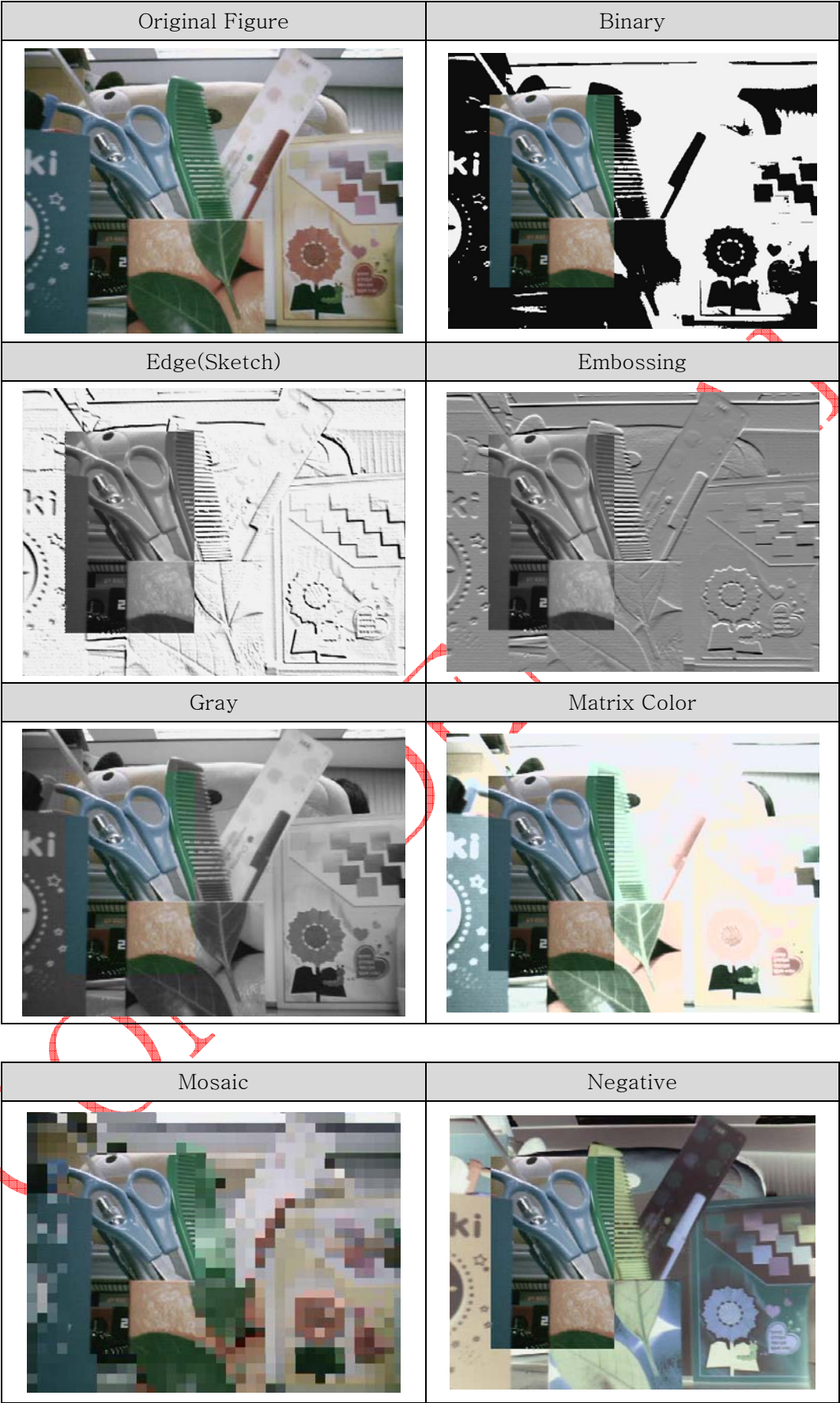


[Figure5-14] Image upscale of NX1005™

5.5.5 Image Effect and Background Image Control

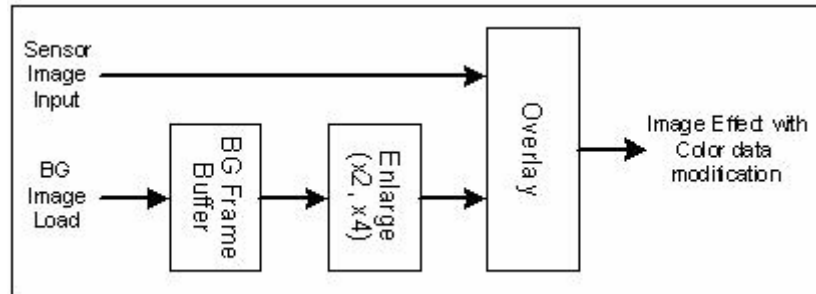
Two kinds of image effect are supported by NX1005™. One is sensor input data modification and the other is synthesis with pre-define back ground image. NX1005™ supports various image effects like EMBOSS, EDGE, NEGATIVE, BINARY, GRAY and MOSAIC.

There are additional categories of image effects, image synthesis with background image like superimpose, overlap.



[Figure 5-15] Sample Figures of Image Effect

For these functions, RGB background image data should be stored in internal memory before function is activated. Image synthesis process between background and sensor input image is as shown below.



[Figure5-16] Image synthesis flow of NX1005™

Overlay logic performs superimpose or overlap functions. Users can use from 16 bits / pixel image to 1 bits / pixel image for BG image. With small BG frame buffer (stacked memory), bigger BG image can be used with less color.



[Figure5-17] Sample Figures of Background

5.5.6 JPEG Encoding and Decoding

NX1005™ supports standard JPEG compression for capturing images. User can change 8 x 8 quality factor table for quantization process. The different Q-table can be applied for Y and UV.

The JPEG image width and height defines the image size compressed by JPEG engine. These image size values should be round up values from divided input image size defined on JPEG scale/window registers. For example, if input image size is 198x140, JPEG image height and width should be programmed as 18 and 13 each. Because there is 8 lines delay from input image data to JPEG compressed data, it's safe to make VSYNC width from sensor longer then 8-line period for full encoding performance.

Because the JPEG algorithm uses DTC and VLC (Variable Length Coding), JPEG file size is not fixed for the same input image size. The host should read JPEG image size first before reading actual image frame buffer. The JPEG images from NX1005™ contain SOS (Start of Scan)

information and EOI (End of Image) information. The images also contain dummy 0x00 byte after 0xff value. User just adds 624 bytes header information to JPEG images from NX1005™ to get final JPEG file.

NX1005™ supports also JPEG decoding. Normal JPEG files can be decoded by using hardware JPEG decoder, and Motion JPEG can be decoded by using time stamp information. NX1005™ has parsers to extract parameters from header, but JPEG images made by NX1005™ does not have header. In that case, some parameters are needed to decode images stored with NX1005™ encoder. While Q-table and image size parameter for JPEG encoder can be downloaded to JPENG decoder, no extra work is required and images can be easily decoded. Recording and play speed can be also controlled by register setting.

5.5.7 LCD Interface and Video Interface

NX1005™ has direct LCD module interface logic for auto preview without host CPU's burden. There is one kind of LCD module interface, parallel CPU interface. Normal small size LCD module has parallel CPU interface, Intel type. For those LCD modules, while continuous address space for display frame shall be used for NX1005™, preview scheme shall be designed carefully. 18bits LCD data are used for data transfer through parallel CPU.

NX1005™ supports up to 3M-pixel resolution and true color LCD. While frame speed is dependent on input sensor frame speed, sensor operation speed should be fast enough for real time display. Normally, 30fps speed is supported by 300K-pixel sensor, and even by 1.3M-pixel sensor, when it has sub sampling function.

5.6 Internal MCU Interface

5.6.1 Using Command Buffers

NX1005™ has 4 registers for protocol communication with external host. If the host needs to control the internal MCU of NX1005™, host writes the data to the CMDDATA (Bank4/reg0xB001) with CMDNUMBER in Bank4/0xB103. Internal MCU of NX1005™ checks write status of register bank4/0xB101, and if there are some command execution requests, then read the command buffers to execute command from the host CPU. Up to 4 sequential commands can be transferred at one time. Internal MCU interprets the commands and performs the proper action. Using these registers host can control sensor or other external device without direct interface to them. To control the sensor, NX1005™ uses I2C bus. To control the other external devices NX1005™ can use 8 GPIO pins. NX1005™ has 8 GPIO pins. It's controlled by firmware of internal MCU. 8 GPIO pins can be controlled by host command through CMD.

GPIO can be used in the applications where some hardware signals are needed like zoom control,

auto focus control, flash light control etc. It's convenient to use NX1005™'s GPIO ports instead of host CPU's GPIO ports. These GPIOs also can be controlled directly through Bank4/reg0x9008 ~ 0x900A. Some times it's convenient for external host controls these GPIO directly. But if some computation is needed to decide GPIO operations, it's better for internal MCU controls GPIOs to minimize load of external host.

For example, if it is required to generate hardware signals simply high or low for moving motor, host can directly control GPIOs. But if some signal patterns are required for motor control, it's convenient for internal MCU to generate output patterns according to host's commands. Actually, the motor for zoom lens control shall be located closer to NX1005™ than to host CPU, so it's better to access zoom motor by NX1005™ than by host.

For flash light control, GPIO also can be used. When one GPIO is connected to switch of flash, the flash can be controlled by NX1005™. The flash can be controlled automatically by internal MCU without any host CPU's burden. Host can control flash operation through GPIO control, but some image analysis are needed normally to decide flash light on/off, so it's convenient to control flash light switching by internal MCU.

5.6.2 Auto-Focusing Control

NX1005™ 3x3-matrix calculation function is to implement of BPF (Band Pass Filter) and edge detection filter. NX1005™ uses only Y values for every pixel to calculate edge values. The summation of BPF and edge value of every sensor frame from this filter is stored in bank4/reg0xC022 ~ 0xC049. User can define edge filter and two windows for AF summation. Internal MCU or external host should read these summation value to decide focus is good or but. If current focus is not good enough, user can control motors for lens moving. As above section, these actions are performed through GPIO control.

While same logic block is used both for auto focus and image effect operation, some of image effect wouldn't be working during AF operation. The window definition for AF summation is also same with image effect.

5.6.3 Firmware Download

NX1005™ has 4KB RAM for internal programming. These RAM can be used to save program to be executed by internal MCU. With this programming, internal MCU can control sensor or execute some algorithm calculation like auto focus. Because most of functions about sensor control are implemented by firmware, firmware programming is very important. With this firmware, flexible operation can be supported. Normally the firmware will be fixed after the determination of what sensor will be used and what algorithm will be used for the sensor control. So we use code RAM instead of ROM for user expansion.

There are two ways to download firmware to code RAM. The program can be loaded from the external serial EPROM or host CPU through parallel port. If NX1005™ finds EPROM on I2C bus, it reads the data into internal RAM. The firmware also can be downloaded through parallel bus from external host. 2 registers Bank4/reg0x9006, 0x9007 are assigned for firmware download from external host through parallel bus. The firmware download sequences are as below.

- A) Enable internal code RAM access. Set STARTUP_CODE_DOWN_EN of Bank4/reg 0x9006.
- B) Write firmware through STARTUP_CODE register – Bank4/reg0x9007. Internal address is automatically increased with sequential data write.
- C) Disable internal code RAM access. Clear STARTUP_CODE_DOWN_EN of Bank4/reg 0x9006.
This will reset the internal MCU for internal MCU to restart with firmware in the RAM.
- D) A~C can be used again to download other firmware.

5.7 PLL Setting

NX1005™ has built-in PLL to generate internal clock and sensor clock. So NX1005™ does not need to have additional clock, but uses any clock already used for main mobile phone operation. The PLL output frequency range is 50MHz ~ 150MHz. The output frequency is defined as below.

$$F_{OUT} = (M * F_{IN}) / (P * 2).$$

Where main-divisor is M[10:0] of Bank4/reg0x9002 , pre-divisor is P[8:0] of Bank4/reg0x09003.

6. Electrical Characteristics

6.1 Absolute Maximum Ratings(VSS=0V)

Characteristics	Symbol	Rating	Unit
Supply Voltage	V _{dd}	-0.3 to 4.5	V
DC Input Voltage	V _{in}	3.8	V
DC Input Current	I _{in}	+/- 200	mA
Storage Temperature	T _{stg}	-65 to 150	C

6.2. Recommended Operating Conditions

Characteristics	Symbol	Min	Typ	Max	Units
IO Voltage	VDDOP	1.62 2.52	1.8 2.8	1.98 3.08	V
Core Voltage	VDDI	1.62	1.8	1.98	V
PLL Voltage	AVDD_PLL	1.62	1.8	1.98	V
Operating Temperature	T _{opr}	-40	25	125	C
Oscillation Freq.	F _{osc}	12	-	30	MHz

6.3. DC Characteristics

Characteristics	Conditions	Symbol	Min	Typ	Max	Units
Input low	VDDOP=2.8V	V _{IL}	-	-	0.35*VDDOP	V
CMOS voltage	VDDOP=1.8V	V _{IL}	-	-	0.3*VDDOP	V
Input high	VDDOP=2.8V	V _{IH}	0.65*VDDOP	-	-	V
CMOS voltage	VDDOP=1.8V	V _{IH}	0.7*VDDOP	-	-	V
Schmitt trigger +	VDDOP=2.8V	V _{t+}	1.51	1.64	1.74	V
	VDDOP=1.8V	V _{t+}	1.02	1.15	1.25	V
Schmitt trigger -	VDDOP=2.8V	V _{t-}	1.01	1.13	1.23	V
	VDDOP=1.8V	V _{t-}	0.64	0.71	0.78	V
Output low Voltage	VDDOP=2.8V	V _{OL}	-	-	0.4	V
	VDDOP=1.8V	V _{OL}	-	-	0.4	V
Output high Voltage	VDDOP=2.8V	V _{OH}	VDDOP-0.4	-	-	V
	VDDOP=1.8V	V _{OH}	VDDOP-0.4	-	-	V

6.4. Power Consumption

< Power save mode >

- POWER DOWN: disable all clocks. SDRAM is deep power down mode.
- IDLE: enable register clock. SDRAM is self refresh mode.
- ACTIVE: enable current functional block clock.
- In POWER DOWN mode, direct LCD control by host.

Parameter	Symbol	Condition	Min	Typ	Max	Unit
*1)Operating Current	Idd1	Vdd = Typ			45	mA
*2)Operating Current	Idd2	Vdd = Typ			60	mA
*3)Operating Current	Idd3	Vdd = Typ			2	mA
*4)Operating Current	Idds	Vdd = Typ			150	μA

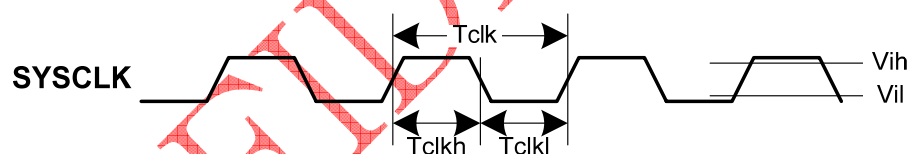
*1) Camera preview operation (48MHz)

*2) 3D graphic operation (54MHz) : about 1mA / MHz

*3) Idle mode

*4) Power down mode (Bypass)

6.5. Internal System Clock Timing



[Figure 6-1] Clock Timing Diagram

Parameter	Symbol	Min	Typ	Max	Unit
Clock Period	Tclk	8.3	-	78	ns
Clock logic high	Tclkh	3	-	35	ns
Clock logic low	TclkL	3	-	35	ns
High level input voltage	Vih	0.8Vdd	-	-	ns
Low level input voltage	Vil	-	-	0.2Vdd	ns

6.6. Recommend Specification of internal PLL

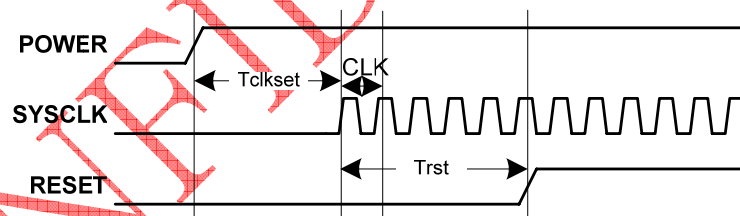
Parameter	Symbol	Min	Typ	Max	Unit
Input Frequency	X_IN	1	-	30	MHz
Jitter	Jrms	-	400	-	ps
	Jpp	-	2	-	ns
Duty Cycle		45	50-	55	%
Locking Time	Tlock	-	-	10	ms
Power Supply	AVDD_PLL	1.7	1.8	1.9	V

[Table 6-1] shows the various external clock, PLL and internal clocks.

Source Clock	PLL Out	System Clock	Graphic Clock	SDRAM Clock	TV-out Clock	Unit
12~30	48	48	48	48	27	MHz
	54	54	54	54		
	81	40.5	81	81		
	96	48	96	96		
	108	54	54	54		
	135	45	67.5	67.5		
	144	48	72	72		

[Table 6-1] Clock Source & Internal Clocks

6.7. Reset Timing



[Figure 6-2] Reset Timing Diagram

Parameter	Symbol	Min	Typ	Max	Unit
Clock set time	Tclkset	300	-	-	μs
Reset period	Trst	CLK * 64	-	-	μs

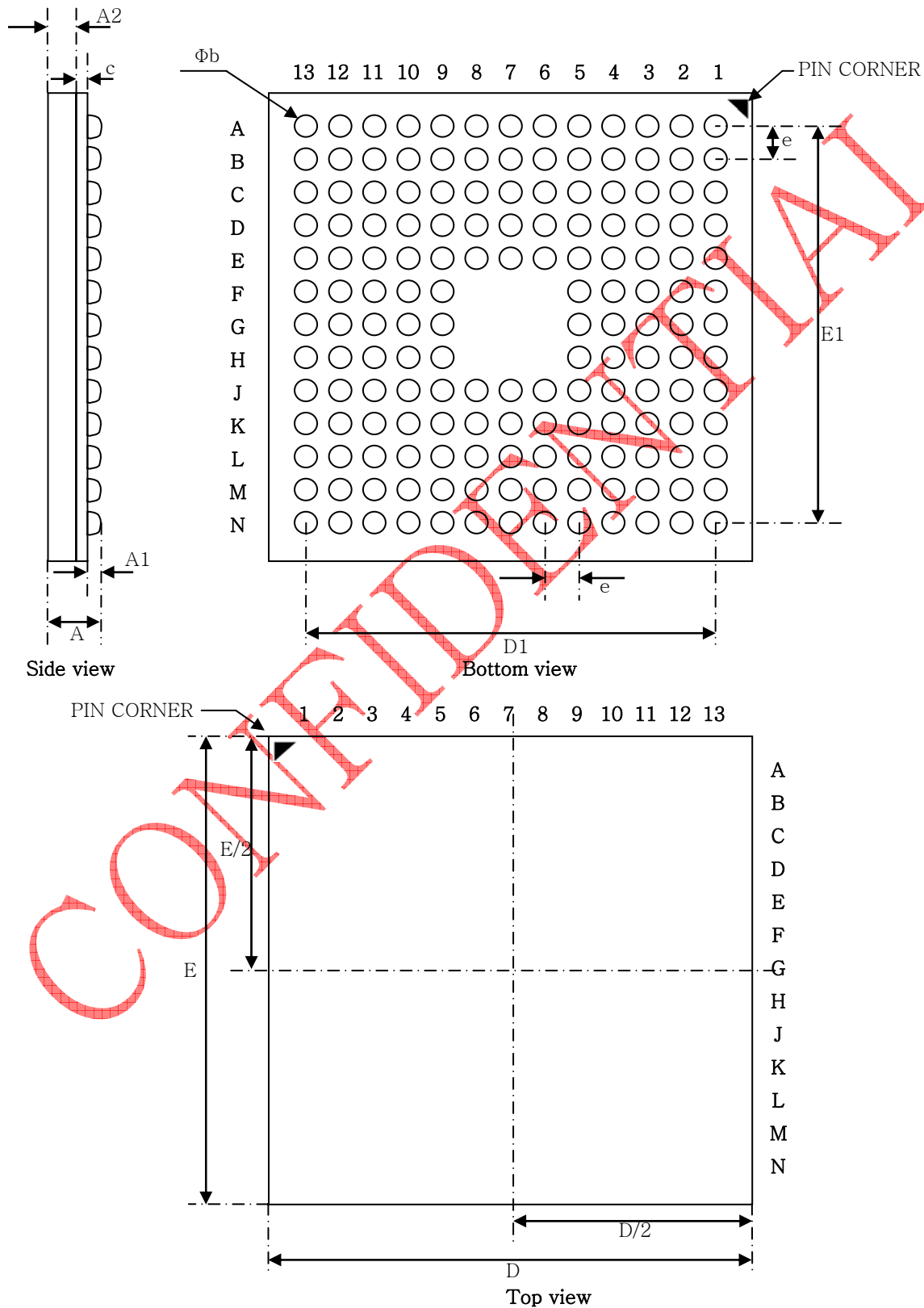
7. Package

7.1 Package Diagram

	1	2	3	4	5	6	7	8	9	10	11	12	13	
A	RST	VDDOP S0	VSS	S_HSYN C	S_SCL	S_DIN0	S_DIN5	GPIO2	GPIO7	NC	L_DATA 16	VDDOP S1	NC	
B	VSS	TMODE 0	TMODE 1	S_VSYN C	S_SDA	S_DIN1	S_DIN6	GPIO3	NC	NC	L_DATA 17	NC	VSS	
C	TMODE 2	M_DA TA0	M_DA TA1	S_MCLK	S_PCLK	S_DIN2	S_DIN7	GPIO4	NC	NC	NC	NC	NC	
D	M_DA TA2	M_DA TA3	M_DA TA4	M_RD	M_INTR	S_DIN3	GPIO0	GPIO5	NC	L_CS_S	L_CS_M	L_RD	L_WR	
E	M_DA TA5	M_DA TA6	M_DA TA7	VSS	VDDI	S_DIN4	GPIO1	GPIO6	VSS	VSS	L_ADS	L_DATA 0	L_DATA 1	
F	M_DA TA8	M_DA TA9	M_DA TA10	M_DA TA11	M_DA TA12	TOP VIEW				L_DATA 2	L_DATA 3	L_DATA 4	L_DATA 5	L_DATA 6
G	M_DA TA13	M_DA TA14	M_DA TA15	M_AD DR0	M_AD DR1					L_DATA 7	L_DATA 8	L_DATA 9	L_DATA 10	L_DATA 11
H	M_AD DR2	M_AD DR3	M_AD DR4	NC	NC					L_DATA 12	L_DATA 13	L_DATA 14	L_DATA 15	L_DATA 18
J	NC	NC	NC	VSS	VDDOP M	S_PWD N	S_RSTN	NC	VDDI	VSS	VDDOP ES0	VSS	L_DATA 19	
K	AVDD_P LL	NC	NC	NC	NC	NC	NC	NC	NC	VSS	NC	NC	L_DATA 20	
L	NC	M_CS	M_WR	NC	NC	NC	NC	NC	SER_CS	NC	NC	L_DATA 21	L_DATA 22	
M	AVSS_P LL	X_OUT	NC	M_DA TA16	NC	NC	NC	NC	SER_CHK	TV_RST	TV_CLK	TV_CSB	VSS	
N	X_IN	VDDOP ES1	M_DA TA17	NC	NC	NC	NC	SER_DI	SER_DO	NC	NC	VDDOP S2	L_DATA 23	

7.2 Package Information

$A = 1.30\text{mm MAX.}$ $A1 = 0.15 \sim 0.26\text{mm}$ $A2 = 0.80\text{mm}$ $C = 0.26\text{mm}$ $\Phi b = 0.30\text{mm}$
 $D = 9.50\text{mm}$ $D1 = 7.80\text{mm}$ $E = 9.50\text{mm}$ $E1 = 7.80\text{mm}$ $e = 0.65\text{mm}$



Appendix – Register Map

1 Camera Register Map

BANK0 (M_ADDR[4:1] == 0x00)					
LCD ADS Control					
Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	0 / Direct	[0] or [15:0]	LCD_ADS_REG	0x0000 or 0x001F	※ When HOST_ADDR[0] is 0, bit[0] means LCD_ADS. CPU can directly control LCD_ADS output using this register when LCD_ADS select (LCD_CTRL[9:8]) = 01b. Writing “1” to this register makes LCD_ADS goes to “1”. Writing “0” to this register makes LCD_ADS goes to “0”.

BANK1 (M_ADDR[4:1] == 0x01)**Main LCD Control**

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	1 / Direct	[15:0] or [7:0]	MAIN_LCD_DATA		Used when LCD control bypass mode. The D[15:0], WR, RD, CS signals are bypassed to LD[15:0], L_WR, L_RD, L_CS_M with this address.

BANK2 (M_ADDR[4:1] == 0x02)**Sub LCD Control**

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	2 / Direct	[15:0] or [7:0]	SUB_LCD_DATA		Used when LCD control bypass mode. The D[15:0], WR, RD, CS signals are bypassed to LD[15:0], L_WR, L_RD, L_CS_S with this address..

BANK3 (M_ADDR[4:1] == 0x03)**Camera Function Control**

Access	Bank/ Data	Bit	Name	Default	Description
W/R	3 / 0x0000	[15:0]	CAM_NOP		All camera operation stop.
W/R	3 / 0x0001	[15:0]	CAM_SENSOR_PREVIEW_START		Starts the image display from sensor on LCD.
W/R	3 / 0x0002	[15:0]	CAM_SENSOR_PREVIEW_STOP		Stops the image display from sensor on LCD.
W/R	3 / 0x0003	[15:0]	CAM_MEMORY_PREVIEW		Displays the image stored in Memory on LCD.
W/R	3 / 0x0004	[15:0]	CAM_BMP_CAPTURE		Captures as the set size in BMP format.
W/R	3 / 0x0005	[15:0]	CAM_JPEG_CAPTURE		Captures as the set size in JPEG format.
W/R	3 / 0x0006	[15:0]	CAM_MJPEG_CAPTURE_START		Starts capture as large as the set size in MJPEG format.
W/R	3 / 0x0007				Reserved
W/R	3 / 0x0008	[15:0]	CAM_MJPEG_CAPTURE_STOP		Stops capture as large as the set size in MJPEG format.
W/R	3 / 0x0009	[15:0]	CAM_JPEG_DECODE		Displays the JPEG image on LCD after Decoding.
W/R	3 / 0x000A	[15:0]	CAM_MJPEG_DECODE_START		Starts decoding as large as the set size in MJPEG format.
W/R	3 / 0x000B				Reserved
W/R	3 / 0x000C	[15:0]	CAM_MJPEG_DECODE_STOP		Stops decoding as large as the set size in MJPEG format.
W/R	3 / 0x000D	[15:0]	CAM_MULTIJPEG_CAPTURE_START		Starts capture a series of JPEG images as many as the number set in advance.
W/R	3 / 0x000E	[15:0]	CAM_MULTIJPEG_CAPTURE_STOP		Stops capture a series of JPEG images as many as the number set in advance.
W/R	3 / 0x000F	[15:0]	CAM_JPEG2BMP		Converts JPEG image to BMP image

W/R	3 / 0x0010	[15:0]	CAM_BMP2JPEG		Converts BMP image to JPEG image
W/R	3 / 0x0011	[15:0]	CAM_YUV_CAPTURE1		Captures as the set size in YUV 422 pixel format.
W/R	3 / 0x0012	[15:0]	CAM_YUV_CAPTURE2		Captures as the set size in YUV 422/420 block format.
W/R	3 / 0x0013	[15:0]	CAM_RGB2YUV		Converts RGB data to YUV data

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BANK4 (M_ADDR[4:1] == 0x04)**Preview Control**

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x0000	[10:0]	PV_H_CROP	0x0000	Preview horizontal crop
W/R	4 / 0x0001	[10:0]	PV_V_CROP	0x0000	Preview vertical crop
W/R	4 / 0x0002	[11:0]	PV_SCALE_SRC_H_SIZ E	0x000	Preview scale source horizontal size
W/R	4 / 0x0003	[11:0]	PV_SCALE_SRC_V_SIZ E	0x000	Preview scale source vertical size
W/R	4 / 0x0004	[11:0]	PV_SCALE_DST_H_SIZ E	0x000	Preview scale destination horizontal size
W/R	4 / 0x0005	[11:0]	PV_SCALE_DST_V_SIZ E	0x000	Preview scale destination vertical size
W/R	4 / 0x0006	[7:0]	PV_SCALE_H_FS_COF F0	0x00	Preview scale horizontal filter coefficient 0
W/R	4 / 0x0007	[7:0]	PV_SCALE_H_FS_COF F1	0x00	Preview scale horizontal filter coefficient 1
W/R	4 / 0x0008	[7:0]	PV_SCALE_H_FS_COF F2	0x00	Preview scale horizontal filter coefficient 2
W/R	4 / 0x0009	[7:0]	PV_SCALE_H_FS_COF F3	0x00	Preview scale horizontal filter coefficient 3
W/R	4 / 0x000A	[7:0]	PV_SCALE_H_FS_COF F4	0x00	Preview scale horizontal filter coefficient 4
W/R	4 / 0x000B	[7:0]	PV_SCALE_H_FS_COF F5	0x00	Preview scale horizontal filter coefficient 5
W/R	4 / 0x000C	[7:0]	PV_SCALE_H_FS_COF F6	0x00	Preview scale horizontal filter coefficient 6
W/R	4 / 0x000D	[7:0]	PV_SCALE_V_FS_COF F0	0x00	Preview scale vertical filter coefficient 0
W/R	4 / 0x000E	[7:0]	PV_SCALE_V_FS_COF F1	0x00	Preview scale vertical filter coefficient 1
W/R	4 / 0x000F	[7:0]	PV_SCALE_V_FS_COF F2	0x00	Preview scale vertical filter coefficient 2

W/R	4 / 0x0010	[7:0]	PV_SCALE_V_FS_COF F3	0x00	Preview scale vertical filter coefficient 3
W/R	4 / 0x0011	[7:0]	PV_SCALE_V_FS_COF F4	0x00	Preview scale vertical filter coefficient 4
W/R	4 / 0x0012	[7:0]	PV_SCALE_V_FS_COF F5	0x00	Preview scale vertical filter coefficient 5
W/R	4 / 0x0013	[7:0]	PV_SCALE_V_FS_COF F6	0x00	Preview scale vertical filter coefficient 6
W/R	4 / 0x0014	[9:0]	PV_START_ADDR0H	0x00	Preview start address 0 high [25:16]
W/R	4 / 0x0015	[15:0]	PV_START_ADDR0L	0x0000	Preview start address 0 low [15:0]
W/R	4 / 0x0016	[9:0]	PV_START_ADDR1H	0x00	Preview start address 1 high [25:16]
W/R	4 / 0x0017	[15:0]	PV_START_ADDR1L	0x0000	Preview start address 1 low [15:0]
W/R	4 / 0x0018				Reserved
W/R	4 / 0x0019	[6:0]	MAIN_LCD_H_CROP	0x00	Main LCD horizontal crop
W/R	4 / 0x001A	[6:0]	MAIN_LCD_V_CROP	0x00	Main LCD vertical crop
W/R	4 / 0x001B	[8:0]	MAIN_LCD_H_SIZE	0x000	Main LCD horizontal size
W/R	4 / 0x001C	[8:0]	MAIN_LCD_V_SIZE	0x000	Main LCD vertical size
W/R	4 / 0x001D	[0]	PV_SCALE_CTRL	0	Preview scale control [0] : Preview LPF enable
W/R	4 / 0x001E	[11:0]	PV_MEM_SRC_H_SIZE	0x000	Preview memory source horizontal size
W/R	4 / 0x001F	[11:0]	PV_MEM_SRC_V_SIZE	0x000	Preview memory source vertical size
W/R	4 / 0x0020	[9:0]	BMP_CAP_START_AD DRH	0x00	BMP capture start address high [25:16]
W/R	4 / 0x0021	[15:0]	BMP_CAP_START_AD DRL	0x0000	BMP capture start address low [15:0]
W/R	4 / 0x0022	[9:0]	YUV_CAP_START_AD DRH	0x00	YUV capture start address high [25:16]

W/R	4 / 0x0023	[15:0]	YUV_CAP_START_AD DRL	0x0000	YUV capture start address low [15:0]
W/R	4 / 0x0024	[6:0]	SUB_LCD_H_CROP	0x00	Sub LCD horizontal crop
W/R	4 / 0x0025	[6:0]	SUB_LCD_V_CROP	0x00	Sub LCD vertical crop
W/R	4 / 0x0026	[5:0]	SUB_LCD_HV_SKIP	0x00	Sub LCD horizontal/vertical skip
W/R	4 / 0x0027	[8:0]	SUB_LCD_H_CSIZE	0x000	Sub LCD horizontal crop size
W/R	4 / 0x0028	[8:0]	SUB_LCD_V_CSIZE	0x000	Sub LCD vertical crop size
W/R	4 / 0x0029	[8:0]	SUB_LCD_H_SIZE	0x000	Sub LCD horizontal size
W/R	4 / 0x002A	[8:0]	SUB_LCD_V_SIZE	0x000	Sub LCD vertical size
W/R	4 / 0x002B	[11:0]	PV_STEREO_WCUT	0x000	Stereo preview mode scaler horizontal crop size
W/R	4 / 0x002C	[10:0]	PV_STEREO_HCUT	0x000	Stereo preview mode scaler vertical crop size
W/R	4 / 0x002D	[11:0]	PV_STEREO_WIDTH	0x000	Stereo preview mode scaler output horizontal size
W/R	4 / 0x002E	[10:0]	PV_STEREO_HEIGHT	0x000	Stereo preview mode scaler output vertical size
W/R	4 / 0x002F	[5:0]	PV_STEREO_SAMPLE	0x000	Stereo preview mode scaler skip size

Capture Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x1000	[10:0]	CAP_H_CROP	0x0000	Capture horizontal crop
W/R	4 / 0x1001	[10:0]	CAP_V_CROP	0x0000	Capture vertical crop
W/R	4 / 0x1002	[11:0]	CAP_SCALE_SRC_H_S IZE	0x000	Capture scale source horizontal size
W/R	4 / 0x1003	[11:0]	CAP_SCALE_SRC_V_S IZE	0x000	Capture scale source vertical size
W/R	4 / 0x1004	[11:0]	CAP_SCALE_DST_H_S IZE	0x000	Capture scale destination horizontal size
W/R	4 / 0x1005	[11:0]	CAP_SCALE_DST_V_S IZE	0x000	Capture scale destination vertical size
W/R	4 / 0x1006	[7:0]	CAP_SCALE_H_FS_CO FF0	0x00	Capture scale horizontal filter coefficient 0
W/R	4 / 0x1007	[7:0]	CAP_SCALE_H_FS_CO FF1	0x00	Capture scale horizontal filter coefficient 1
W/R	4 / 0x1008	[7:0]	CAP_SCALE_H_FS_CO FF2	0x00	Capture scale horizontal filter coefficient 2
W/R	4 / 0x1009	[7:0]	CAP_SCALE_H_FS_CO FF3	0x00	Capture scale horizontal filter coefficient 3
W/R	4 / 0x100A	[7:0]	CAP_SCALE_H_FS_CO FF4	0x00	Capture scale horizontal filter coefficient 4
W/R	4 / 0x100B	[7:0]	CAP_SCALE_H_FS_CO FF5	0x00	Capture scale horizontal filter coefficient 5
W/R	4 / 0x100C	[7:0]	CAP_SCALE_H_FS_CO FF6	0x00	Capture scale horizontal filter coefficient 6
W/R	4 / 0x100D	[7:0]	CAP_SCALE_V_FS_CO FF0	0x00	Capture scale vertical filter coefficient 0
W/R	4 / 0x100E	[7:0]	CAP_SCALE_V_FS_CO FF1	0x00	Capture scale vertical filter coefficient 1
W/R	4 / 0x100F	[7:0]	CAP_SCALE_V_FS_CO FF2	0x00	Capture scale vertical filter coefficient 2
W/R	4 / 0x1010	[7:0]	CAP_SCALE_V_FS_CO FF3	0x00	Capture scale vertical filter coefficient 3

W/R	4 / 0x1011	[7:0]	CAP_SCALE_V_FS_CO FF4	0x00	Capture scale vertical filter coefficient 4
W/R	4 / 0x1012	[7:0]	CAP_SCALE_V_FS_CO FF5	0x00	Capture scale vertical filter coefficient 5
W/R	4 / 0x1013	[7:0]	CAP_SCALE_V_FS_CO FF6	0x00	Capture scale vertical filter coefficient 6
W/R	4 / 0x1014	[9:0]	CAP_SRC_START_ADD RH	0x00	Capture source start address high [25:16]
W/R	4 / 0x1015	[15:0]	CAP_SRC_START_ADD RL	0x0000	Capture source start address low [15:0]
W/R	4 / 0x1016	[9:0]	CAP_DST_START_ADD RH	0x00	Capture destination start address high [25:16]
W/R	4 / 0x1017	[15:0]	CAP_DST_START_ADD RL	0x0000	Capture destination start address low [15:0]
W/R	4 / 0x1018	[7:0]	CAP_Q_FACTOR_DAT A	0x00	Capture quality factor data Internal 64x8 bits quality factor RAM can be updated through this register access.
W/R	4 / 0x1019	[1:0]	CAP_Q_FACTOR_CTRL	0x0	Capture quality factor control [1] : Quality factor use 0 : JPEG codec access Q-table 1 : Host access Q-table [0] : Quality factor select 0 : Select luminance quality factor 1 : Select chrominance quality factor
W/R	4 / 0x101A	[7:0]	CAP_JENC_SRC_H_SIZ E	0x00	JPEG encoder source horizontal size (= JPEG image horizontal size / 16)
W/R	4 / 0x101B	[8:0]	CAP_JENC_SRC_V_SIZ E	0x000	JPEG encoder source vertical size (= JPEG image vertical size / 8 : YUV422) (= JPEG image vertical size / 16 : YUV420)
W/R	4 / 0x101C	[10:0]	CAP_JENC_SRC_H_SIZ E_LIMIT	0x000	JPEG encoder source horizontal size limit (= JPEG image horizontal size / 2)
W/R	4 / 0x101D	[11:0]	CAP_MEM_SRC_H_SIZ E	0x000	Capture memory source horizontal size
W/R	4 / 0x101E	[11:0]	CAP_MEM_SRC_V_SIZ E	0x000	Capture memory source vertical size

W/R	4 / 0x101F	[1:0]	CAP_MEM_SRC_FORM AT	0x0	Capture memory source format 0 : RGB565 format (16 bit) 1 : RGB666 format (2 + 16 bit) 2 : RGB888 format (8 + 16 bit) 3 : Reserved
W/R	4 / 0x1020	[1:0]	CAP_MEM_DST_FORM AT	0x0	Capture memory destination format 0 : RGB565 format (16 bit) 1 : RGB666 format (2 + 16 bit) 2 : RGB888 format (8 + 16 bit) 3 : Reserved
W/R	4 / 0x1021	[0]	CAP_SCALE_CTRL	0	Capture scale control [0] : Capture LPF enable
R	4 / 0x1022	[15:0]	CAP_FILE_SIZEH	0x00	Capture file size high [31:16]
R	4 / 0x1023	[15:0]	CAP_FILE_SIZEL	0x0000	Capture file size low [15:0]
W/R	4 / 0x1024	[1:0]	JENC_MODE	0x0	JPEG encoder mode 0 : No JPEG encoding action 1 : Single-shot 2 : Multi-shot 3 : Motion-JPEG
W/R	4 / 0x1025	[0]	JENC_FORMAT	0	JPEG encoder format 0 : YUV422 format 1 : YUV420 format
W/R	4 / 0x1026	[0]	CAP_MEM_ADDR_MO DE	0	Capture memory address mode 0 : Normal address mode 1 : Matrix address mode
W/R	4 / 0x1027	[11:0]	CAP_STEREO_WCUT	0x000	Stereo capture mode scaler horizontal crop size
W/R	4 / 0x1028	[10:0]	CAP_STEREO_HCUT	0x000	Stereo capture mode scaler vertical crop size
W/R	4 / 0x1029	[11:0]	CAP_STEREO_WIDTH	0x000	Stereo capture mode scaler output horizontal size
W/R	4 / 0x102A	[10:0]	CAP_STEREO_HEIGHT	0x000	Stereo capture mode scaler output vertical size
W/R	4 / 0x102B	[5:0]	CAP_STEREO_SAMPLE	0x000	Stereo capture mode scaler skip size

Decoding Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x2000	[10:0]	DEC_H_CROP	0x0000	Decoding horizontal crop
W/R	4 / 0x2001	[10:0]	DEC_V_CROP	0x0000	Decoding vertical crop
W/R	4 / 0x2002	[11:0]	DEC_SCALE_SRC_H_S IZE	0x000	Decoding scale source horizontal size
W/R	4 / 0x2003	[11:0]	DEC_SCALE_SRC_V_S IZE	0x000	Decoding scale source vertical size
W/R	4 / 0x2004	[11:0]	DEC_SCALE_DST_H_S IZE	0x000	Decoding scale destination horizontal size
W/R	4 / 0x2005	[11:0]	DEC_SCALE_DST_V_S IZE	0x000	Decoding scale destination vertical size
W/R	4 / 0x2006	[7:0]	DEC_SCALE_H_FS_CO FF0	0x00	Decoding scale horizontal filter coefficient 0
W/R	4 / 0x2007	[7:0]	DEC_SCALE_H_FS_CO FF1	0x00	Decoding scale horizontal filter coefficient 1
W/R	4 / 0x2008	[7:0]	DEC_SCALE_H_FS_CO FF2	0x00	Decoding scale horizontal filter coefficient 2
W/R	4 / 0x2009	[7:0]	DEC_SCALE_H_FS_CO FF3	0x00	Decoding scale horizontal filter coefficient 3
W/R	4 / 0x200A	[7:0]	DEC_SCALE_H_FS_CO FF4	0x00	Decoding scale horizontal filter coefficient 4
W/R	4 / 0x200B	[7:0]	DEC_SCALE_H_FS_CO FF5	0x00	Decoding scale horizontal filter coefficient 5
W/R	4 / 0x200C	[7:0]	DEC_SCALE_H_FS_CO FF6	0x00	Decoding scale horizontal filter coefficient 6
W/R	4 / 0x200D	[7:0]	DEC_SCALE_V_FS_CO FF0	0x00	Decoding scale vertical filter coefficient 0
W/R	4 / 0x200E	[7:0]	DEC_SCALE_V_FS_CO FF1	0x00	Decoding scale vertical filter coefficient 1
W/R	4 / 0x200F	[7:0]	DEC_SCALE_V_FS_CO FF2	0x00	Decoding scale vertical filter coefficient 2
W/R	4 / 0x2010	[7:0]	DEC_SCALE_V_FS_CO FF3	0x00	Decoding scale vertical filter coefficient 3

W/R	4 / 0x2011	[7:0]	DEC_SCALE_V_FS_CO FF4	0x00	Decoding scale vertical filter coefficient 4
W/R	4 / 0x2012	[7:0]	DEC_SCALE_V_FS_CO FF5	0x00	Decoding scale vertical filter coefficient 5
W/R	4 / 0x2013	[7:0]	DEC_SCALE_V_FS_CO FF6	0x00	Decoding scale vertical filter coefficient 6
W/R	4 / 0x2014	[9:0]	DEC_SRC_START_ADD RH	0x00	Decoding source start address high [25:16]
W/R	4 / 0x2015	[15:0]	DEC_SRC_START_ADD RL	0x0000	Decoding source start address low [15:0]
W/R	4 / 0x2016	[9:0]	DEC_DST_START_ADD RH	0x00	Decoding destination start address high [25:16]
W/R	4 / 0x2017	[15:0]	DEC_DST_START_ADD RL	0x0000	Decoding destination start address low [15:0]
W/R	4 / 0x2018	[0]	DEC_PARA_UP	0	Decoding parameter up
W/R	4 / 0x2019	[0]	DEC_SCALE_CTRL	0	Decoding scale control [0] : Decoding LPF enable
W/R	4 / 0x201A	[11:0]	DEC_JDEC_H_SIZE	0x000	JPEG decoder target horizontal size
W/R	4 / 0x201B	[11:0]	DEC_JDEC_V_SIZE	0x000	JPEG decoder target vertical size
W/R	4 / 0x201C	[0]	JDEC_MODE	0	JPEG decoder mode 0 : Single/Multi-shot 1 : Motion-JPEG
W/R	4 / 0x201D	[1:0]	DEC_MEM_DST_FOR MAT	0x0	Decoding memory destination format 0 : RGB565 format (16 bit) 1 : RGB666 format (2 + 16 bit) 2 : RGB888 format (8 + 16 bit) 3 : YUV
W/R	4 / 0x201E	[3:0]	DEC_ERROR_STATUS	0x0	Decoding error status [3] : Memory overflow [2] : Unpredictable data error [1] : Unsupported JPEG mode and Huffman table error [0] : Unsupported YUV format error

W/R	4 / 0x201F	[2:0]	YUV_DST_MODE	0x0	YUV destination format 7 : Reserved 6 : Block and white 5 : YUV411V 4 : YUV411H 3 : YUV420 2 : YUV422V 1 : YUV422H 0 : YUV444
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Zoom Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x3000	[15:0]	ZOOM_CTRL	0x0	Zoom control [15:8] : Zoom weight for up-scaling This should be $\{1280 - (N*1280)/M\} / 10$ [7:1] : Reserved [0] : Zoom enable
W/R	4 / 0x3001	[8:0]	ZOOM_M_FACTOR	0x000	Zoom M factor for M/N scaling
W/R	4 / 0x3002	[8:0]	ZOOM_N_FACTOR	0x000	Zoom N factor for M/N scaling
W/R	4 / 0x3003	[11:0]	ZOOM_SRC_H_SIZE	0x000	Zoom source horizontal size
W/R	4 / 0x3004	[11:0]	ZOOM_SRC_V_SIZE	0x000	Zoom source vertical size
W/R	4 / 0x3005	[11:0]	ZOOM_DST_H_SIZE	0x000	Zoom destination horizontal size
W/R	4 / 0x3006	[11:0]	ZOOM_DST_V_SIZE	0x000	Zoom destination vertical size
W/R	4 / 0x3007	[11:0]	ZOOM_CLIP_START_H_POINT	0x000	Zoom clip start horizontal point
W/R	4 / 0x3008	[11:0]	ZOOM_CLIP_END_H_POINT	0x000	Zoom clip end horizontal point
W/R	4 / 0x3009	[11:0]	ZOOM_CLIP_START_V_POINT	0x000	Zoom clip start vertical point
W/R	4 / 0x300A	[11:0]	ZOOM_CLIP_END_V_POINT	0x000	Zoom clip end vertical point

MJPEG Control

Access	Bank/ Sub-Address	Bit	Name	Default	Description
W/R	4 / 0x4000	[15:0]	FRAME_COUNT	0x00	Target number of frames to be captured in multi-shot or MJPEG encoding
W/R	4 / 0x4001	[1:0]	STOP_MODE	0x0	Stop mode on multi-shot or MJPEG encoding 0 : Stop command 1 : Reserved 2 : Frame count match 3 : Time stamp match
R	4 / 0x4002	[15:0]	FRAME_NUMBER	0x0000	Used to record the number of frame that encoded when multi-shot or MJPEG is stopped
W/R	4 / 0x4003	[15:0]	MS_CLK_COUNT	0x0000	1ms clock count value for MJPEG encoding
W/R	4 / 0x4004	[15:0]	SEC_LIMIT	0x0000	Second limitation for MJPEG encoding
W/R	4 / 0x4005	[15:0]	MS_LIMIT	0x0000	Millisecond limitation for MJPEG encoding
W/R	4 / 0x4006	[9:0]	DSCP_START_ADD RH	0x00	Descriptor start address high [25:16]
W/R	4 / 0x4007	[15:0]	DSCP_START_ADD RL	0x0000	Descriptor start address low [15:0]
W/R	4 / 0x4008	[7:0]	DEC_START_SEC	0x000	Start second for MJPEG decoding
W/R	4 / 0x4009	[9:0]	DEC_START_MS	0x000	Start millisecond for MJPEG decoding
W/R	4 / 0x400A	[2:0]	DEC_SPEED	0x0	Replay speed for MJPEG decoding 0 : Normal speed decoding 1 : 2x speed decoding 2 : 4x speed decoding 3 : Reserved 4 : 1/2x speed decoding 5 : 1/4x speed decoding 6 : Reserved 7 : Reserved
W/R	4 / 0x400B	[15:0]	DEC_DONE_FRAME _NUMBER	0x0000	Decoding done frame number for MJPEG decoding

Sensor Interface Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x5000	[11:0]	SENSOR_CLK_CTRL	0x000	Sensor clock control [11:8] : TVCLK select [11] : TVCLK select 0 : External clock 1 : PLL output clock [10:8] : TVCLK divide 0 : PLL output clock bypass 1 : PLL output clock / 2 2 : PLL output clock / 3 3 : PLL output clock / 4 4 : PLL output clock / 5 5 : PLL output clock / 6 6 : PLL output clock / 7 7 : PLL output clock / 8 [7:6] : Reserved [5:4] : PCLK select 0 : Sensor PCLK 1 : Inverse sensor PCLK 2 : MCLK loopback 3 : MCLK loopback / 2 [3] : MCLK enable 0 : MCLK goes to "0" 1 : MCLK operate normally [2] : Reserved [1:0] : MCLK divide 0 : CAMCLK / 2 1 : CAMCLK 2 : CAMCLK / 4 3 : CAMCLK

W/R	4 / 0x5001	[15:0]	SENSOR_IF_CTRL	0x0002	Sensor interface control [15] : CbCr or UV select 0 : UV select 1 : CbCr select [14] : YUV sample control [13] : YUV data sample point [12] : VSYNC Cut [11] : HSYNC Cut [10] : Inverse VSYNC (Internal VSYNC is low active) 0 : Use input VSYNC polarity 1 : Inversed input VSYNC polarity [9] : Inverse HSYNC (Internal HSYNC is high active) 0 : Use input HSYNC polarity 1 : Inversed input HSYNC polarity [8] : Inversed PCLK (Internal PCLK is high active) 0 : Use input PCLK polarity 1 : Inversed input PCLK polarity [7:4] : Frame rate [3:2] : Color select pattern [1:0] : Sensor pattern 0 : YCbYCr 1 : CbYCrY 2 : YCrYCb 3 : CrYCbY
W/R	4 / 0x5002	[0]	SENSOR_RST	0	Sensor reset This bit is used to reset the sensor using external pin control. Writing “1” to this register makes S_RST pin goes to “1”. Writing “0” to this register makes S_RST pin goes to “0”.
W/R	4 / 0x5003	[7:0]	HV_SYNC_DELAY	0x0036	HSYNC, VSYNC delay [7:4] : VSYNC delay [3:0] : HSYNC delay

W/R	4 / 0x5004	[4:0]	STEREO_CTRL	0x0002	[4] : MIX_RC Whether L sensor or R sensor 0 : LRLR..... 1 : RLRL.... [3] : FRAME_MASK_SEL 0 : L 1 : R [2] : STEREO_ONOFF 0 : Off 1 : On [1] : MIX_WAY Selection using even pixel value 0 : 00224466.... 1 : 01234567... [0] : MIX_V 0 : 1 line—R0L1R2L3R4L5... 2 line—L0R1L2R3L4R5... 1 : 1 line—L0R1L2R3L4R5... 2 line—R0L1R2L3R4L5...
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LCD Interface Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x6000	[0]	LCD_OWN_SEL	0	LCD ownership control 1 : Graphic display 0 : Camera display

W/R	4 / 0x6001	[14:0]	LCD_CTRL	0x0000	LCD control [14] : LCD toggle mode 2 select (Main LCD : sensor data, Sub LCD : host data) 0 : Not select 1 : Select [13:11] : LCD RGB order 7 : Reserved 6 : Reserved 5 : BRG 4 : GBR 3 : GRB 2 : RBG 1 : BGR 0 : RGB [10] : SUB_LCD_CS select 0 : GPIO6 is used as general port IO. 1 : GPIO6 is used as “SUB_LCD_CS” input from host. [9:8] : LCD_ADS select 2, 3 : LCD_ADS pin is controlled by GPIO5. GPIO5 pin is used as “ADS” input from host. 1 : LCD_ADS pin is controlled by register “LCD_ADS_REG” of bank 0. 0 : LCD_ADS pin is controlled by “ADD[0]” input from host. [7] : Special data transfer (2 pixel data 3 phase transfer) 1 : Special data transfer enable 0 : Special data transfer disable [6] : Display write mode 1 : Fast write mode 0 : Normal write mode [5:4] : Display mode 3 : Reserved 2 : TV out 1 : CPU-type display 0 : Host direct display [3] : Main LCD RS signal polarity 1 : High Level 0 : Low level
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					[3] : Main LCD RS signal polarity 1 : High Level 0 : Low level [2] : Sub LCD RS signal polarity 1 : High Level 0 : Low level [1:0] : LCD mode select 3 : Dual 2 : Toggle 1 : Sub LCD only 0 : Main LCD only
W/R	4 / 0x6002	[7:0]	LCD_AUTOCMD_NUM	0x00	LCD auto command number [7:4] : Main LCD number of auto command [3:0] : Sub LCD number of auto command
W/R	4 / 0x6003	[11:0]	LCD_TIMING_CTRL	0x521	LCD timing control (clock cycle base) [11:10] : LCD cs-wr setup time [9:8] : LCD cs-wr hold time [7:4] : LCD wr high period [3:0] : LCD wr low period

W/R	4 / 0x6004	[5:0]	LCD_PHASE_NUM	0x00	LCD preview control 3 [5] : Main LCD single auto command transfer Time 1 : 2 phase transfer 0 : 1 phase transfer [4] : Sub LCD single auto command transfer Time 1 : 2 phase transfer 0 : 1 phase transfer [3:2] : Main LCD single display data transfer time 3: Reserved 2 : 3 phase transfer 1 : 2 phase transfer 0 : 1 phase transfer [1:0] : Sub LCD single display data transfer time 3: Reserved 2 : 3 phase transfer 1 : 2 phase transfer 0 : 1 phase transfer
W/R	4 / 0x6005	[9:0]	MAIN_ADS	0x000	Main LCD auto command ADS level Ex) ADS[0] : 1 st auto cmd ADS level ADS[9] : 10 th auto cmd ADS level
W/R	4 / 0x6006	[9:0]	SUB_ADS	0x000	Sub LCD auto command ADS level Ex) ADS[0] : 1st auto cmd ADS level ADS[9] : 10th auto cmd ADS level
W/R	4 / 0x6007	[15:0]	MAIN_ADDR_DATA0	0x0000	Main LCD 1 st auto command addr/data
W/R	4 / 0x6008	[15:0]	MAIN_ADDR_DATA1	0x0000	Main LCD 2 nd auto command addr/data
W/R	4 / 0x6009	[15:0]	MAIN_ADDR_DATA2	0x0000	Main LCD 3 rd auto command addr/data
W/R	4 / 0x600A	[15:0]	MAIN_ADDR_DATA3	0x0000	Main LCD 4 th auto command addr/data
W/R	4 / 0x600B	[15:0]	MAIN_ADDR_DATA4	0x0000	Main LCD 5 th auto command addr/data
W/R	4 / 0x600C	[15:0]	MAIN_ADDR_DATA5	0x0000	Main LCD 6 th auto command addr/data

W/R	4 / 0x600D	[15:0]	MAIN_ADDR_DATA6	0x0000	Main LCD 7 th auto command addr/data
W/R	4 / 0x600E	[15:0]	MAIN_ADDR_DATA7	0x0000	Main LCD 8 th auto command addr/data
W/R	4 / 0x600F	[15:0]	MAIN_ADDR_DATA8	0x0000	Main LCD 9 th auto command addr/data
W/R	4 / 0x6010	[15:0]	MAIN_ADDR_DATA9	0x0000	Main LCD 10 th auto command addr/data
W/R	4 / 0x6011	[15:0]	SUB_ADDR_DATA0	0x0000	Sub LCD 1 st auto command addr/data
W/R	4 / 0x6012	[15:0]	SUB_ADDR_DATA1	0x0000	Sub LCD 2 nd auto command addr/data
W/R	4 / 0x6013	[15:0]	SUB_ADDR_DATA2	0x0000	Sub LCD 3 rd auto command addr/data
W/R	4 / 0x6014	[15:0]	SUB_ADDR_DATA3	0x0000	Sub LCD 4 th auto command addr/data
W/R	4 / 0x6015	[15:0]	SUB_ADDR_DATA4	0x0000	Sub LCD 5 th auto command addr/data
W/R	4 / 0x6016	[15:0]	SUB_ADDR_DATA5	0x0000	Sub LCD 6 th auto command addr/data
W/R	4 / 0x6017	[15:0]	SUB_ADDR_DATA6	0x0000	Sub LCD 7 th auto command addr/data
W/R	4 / 0x6018	[15:0]	SUB_ADDR_DATA7	0x0000	Sub LCD 8 th auto command addr/data
W/R	4 / 0x6019	[15:0]	SUB_ADDR_DATA8	0x0000	Sub LCD 9 th auto command addr/data
W/R	4 / 0x601A	[15:0]	SUB_ADDR_DATA9	0x0000	Sub LCD 10 th auto command addr/data
W/R	4 / 0x601B	[14:0]	MAIN_AUTOPHASE0_ 23_22_21	0x5ed5	Main LCD 1 st phase auto command bit No.23/22/21 position
W/R	4 / 0x601C	[14:0]	MAIN_AUTOPHASE0_ 20_19_18	0x5272	Main LCD 1 st phase auto command bit No.20/19/18 position
W/R	4 / 0x601D	[14:0]	MAIN_AUTOPHASE0_ 17_16_15	0x460f	Main LCD 1 st phase auto command bit No.17/16/15 position
W/R	4 / 0x601E	[14:0]	MAIN_AUTOPHASE0_ 14_13_12	0x39ac	Main LCD 1 st phase auto command bit No.14/13/12 position
W/R	4 / 0x601F	[14:0]	MAIN_AUTOPHASE0_1 1_10_9	0x2d49	Main LCD 1 st phase auto command bit No.11/10/9 position

W/R	4 / 0x6020	[14:0]	MAIN_AUTOPHASE0_ 8_7_6	0x20e6	Main LCD 1 st phase auto command bit No.8/7/6 position
W/R	4 / 0x6021	[14:0]	MAIN_AUTOPHASE0_ 5_4_3	0x1483	Main LCD 1 st phase auto command bit No.5/4/3 position
W/R	4 / 0x6022	[14:0]	MAIN_AUTOPHASE0_ 2_1_0	0x0820	Main LCD 1 st phase auto command bit No.2/1/0 position
W/R	4 / 0x6023	[14:0]	MAIN_AUTOPHASE1_ 23_22_21	0x5ed5	Main LCD 2 nd phase auto command bit No.23/22/21 position
W/R	4 / 0x6024	[14:0]	MAIN_AUTOPHASE1_ 20_19_18	0x5272	Main LCD 2 nd phase auto command bit No.20/19/18 position
W/R	4 / 0x6025	[14:0]	MAIN_AUTOPHASE1_ 17_16_15	0x460f	Main LCD 2 nd phase auto command bit No.17/16/15 position
W/R	4 / 0x6026	[14:0]	MAIN_AUTOPHASE1_ 14_13_12	0x39ac	Main LCD 2 nd phase auto command bit No.14/13/12 position
W/R	4 / 0x6027	[14:0]	MAIN_AUTOPHASE1_ 11_10_9	0x2d49	Main LCD 2 nd phase auto command bit No.11/10/9 position
W/R	4 / 0x6028	[14:0]	MAIN_AUTOPHASE1_ 8_7_6	0x20e6	Main LCD 2 nd phase auto command bit No.8/7/6 position
W/R	4 / 0x6029	[14:0]	MAIN_AUTOPHASE1_ 5_4_3	0x1483	Main LCD 2 nd phase auto command bit No.5/4/3 position
W/R	4 / 0x602A	[14:0]	MAIN_AUTOPHASE1_ 2_1_0	0x0820	Main LCD 2 nd phase auto command bit No.2/1/0 position
W/R	4 / 0x602B	[14:0]	SUB_AUTOPHASE0_23 _22_21	0x5ed5	Sub LCD 1 st phase auto command bit No.23/22/21 position
W/R	4 / 0x602C	[14:0]	SUB_AUTOPHASE0_20 _19_18	0x5272	Sub LCD 1 st phase auto command bit No.20/19/18 position
W/R	4 / 0x602D	[14:0]	SUB_AUTOPHASE0_17 _16_15	0x460f	Sub LCD 1 st phase auto command bit No.17/16/15 position
W/R	4 / 0x602E	[14:0]	SUB_AUTOPHASE0_14 _13_12	0x39ac	Sub LCD 1 st phase auto command bit No.14/13/12 position
W/R	4 / 0x602F	[14:0]	SUB_AUTOPHASE0_11 _10_9	0x2d49	Sub LCD 1 st phase auto command bit No.11/10/9 position
W/R	4 / 0x6030	[14:0]	SUB_AUTOPHASE0_8_ 7_6	0x20e6	Sub LCD 1 st phase auto command bit No.8/7/6 position
W/R	4 / 0x6031	[14:0]	SUB_AUTOPHASE0_5_ 4_3	0x1483	Sub LCD 1 st phase auto command bit No.5/4/3 position
W/R	4 / 0x6032	[14:0]	SUB_AUTOPHASE0_2_ 1_0	0x0820	Sub LCD 1 st phase auto command bit No.2/1/0 position

W/R	4 / 0x6033	[14:0]	SUB_AUTOPHASE1_23 _22_21	0x5ed5	Sub LCD 2 nd phase auto command bit No.23/22/21 position
W/R	4 / 0x6034	[14:0]	SUB_AUTOPHASE1_20 _19_18	0x5272	Sub LCD 2 nd phase auto command bit No.20/19/18 position
W/R	4 / 0x6035	[14:0]	SUB_AUTOPHASE1_17 _16_15	0x460f	Sub LCD 2 nd phase auto command bit No.17/16/15 position
W/R	4 / 0x6036	[14:0]	SUB_AUTOPHASE1_14 _13_12	0x39ac	Sub LCD 2 nd phase auto command bit No.14/13/12 position
W/R	4 / 0x6037	[14:0]	SUB_AUTOPHASE1_11 _10_9	0x2d49	Sub LCD 2 nd phase auto command bit No.11/10/9 position
W/R	4 / 0x6038	[14:0]	SUB_AUTOPHASE1_8_ 7_6	0x20e6	Sub LCD 2 nd phase auto command bit No.8/7/6 position
W/R	4 / 0x6039	[14:0]	SUB_AUTOPHASE1_5_ 4_3	0x1483	Sub LCD 2 nd phase auto command bit No.5/4/3 position
W/R	4 / 0x603A	[14:0]	SUB_AUTOPHASE1_2_ 1_0	0x0820	Sub LCD 2 nd phase auto command bit No.2/1/0 position
W/R	4 / 0x603B	[14:0]	MAIN_DATAPHASE0_2 3_22_21	0x5ed5	Main LCD 1 st phase data bit No.23/22/21 position
W/R	4 / 0x603C	[14:0]	MAIN_DATAPHASE0_2 0_19_18	0x5272	Main LCD 1 st phase data bit No.20/19/18 position
W/R	4 / 0x603D	[14:0]	MAIN_DATAPHASE0_1 7_16_15	0x460f	Main LCD 1 st phase data bit No.17/16/15 position
W/R	4 / 0x603E	[14:0]	MAIN_DATAPHASE0_1 4_13_12	0x39ac	Main LCD 1 st phase data bit No.14/13/12 position
W/R	4 / 0x603F	[14:0]	MAIN_DATAPHASE0_1 1_10_9	0x2d49	Main LCD 1 st phase data bit No.11/10/9 position
W/R	4 / 0x6040	[14:0]	MAIN_DATAPHASE0_8 _7_6	0x20e6	Main LCD 1 st phase data bit No.8/7/6 position
W/R	4 / 0x6041	[14:0]	MAIN_DATAPHASE0_5 _4_3	0x1483	Main LCD 1 st phase data bit No.5/4/3 position
W/R	4 / 0x6042	[14:0]	MAIN_DATAPHASE0_2 _1_0	0x0820	Main LCD 1 st phase data bit No.2/1/0 position
W/R	4 / 0x6043	[14:0]	MAIN_DATAPHASE1_2 3_22_21	0x5ed5	Main LCD 2 nd phase data bit No.23/22/21 position
W/R	4 / 0x6044	[14:0]	MAIN_DATAPHASE1_2 0_19_18	0x5272	Main LCD 2 nd phase data bit No.20/19/18 position
W/R	4 / 0x6045	[14:0]	MAIN_DATAPHASE1_1 7_16_15	0x460f	Main LCD 2 nd phase data bit No.17/16/15 position

W/R	4 / 0x6046	[14:0]	MAIN_DATAPHASE1_1 4_13_12	0x39ac	Main LCD 2 nd phase data bit No.14/13/12 position
W/R	4 / 0x6047	[14:0]	MAIN_DATAPHASE1_1 1_10_9	0x2d49	Main LCD 2 nd phase data bit No.11/10/9 position
W/R	4 / 0x6048	[14:0]	MAIN_DATAPHASE1_8 _7_6	0x20e6	Main LCD 2 nd phase data bit No.8/7/6 position
W/R	4 / 0x6049	[14:0]	MAIN_DATAPHASE1_5 _4_3	0x1483	Main LCD 2 nd phase data bit No.5/4/3 position
W/R	4 / 0x604A	[14:0]	MAIN_DATAPHASE1_2 _1_0	0x0820	Main LCD 2 nd phase data bit No.2/1/0 position
W/R	4 / 0x604B	[14:0]	MAIN_DATAPHASE2_2 3_22_21	0x5ed5	Main LCD 3 rd phase data bit No.23/22/21 position
W/R	4 / 0x604C	[14:0]	MAIN_DATAPHASE2_2 0_19_18	0x5272	Main LCD 3 rd phase data bit No.20/19/18 position
W/R	4 / 0x604D	[14:0]	MAIN_DATAPHASE2_1 7_16_15	0x460f	Main LCD 3 rd phase data bit No.17/16/15 position
W/R	4 / 0x604E	[14:0]	MAIN_DATAPHASE2_1 4_13_12	0x39ac	Main LCD 3 rd phase data bit No.14/13/12 position
W/R	4 / 0x604F	[14:0]	MAIN_DATAPHASE2_1 1_10_9	0x2d49	Main LCD 3 rd phase data bit No.11/10/9 position
W/R	4 / 0x6050	[14:0]	MAIN_DATAPHASE2_8 _7_6	0x20e6	Main LCD 3 rd phase data bit No.8/7/6 position
W/R	4 / 0x6051	[14:0]	MAIN_DATAPHASE2_5 _4_3	0x1483	Main LCD 3 rd phase data bit No.5/4/3 position
W/R	4 / 0x6052	[14:0]	MAIN_DATAPHASE2_2 _1_0	0x0820	Main LCD 3 rd phase data bit No.2/1/0 position
W/R	4 / 0x6053	[14:0]	SUB_DATAPHASE0_23 _22_21	0x5ed5	Sub LCD 1 st phase data bit No.23/22/21 position
W/R	4 / 0x6054	[14:0]	SUB_DATAPHASE0_20 _19_18	0x5272	Sub LCD 1 st phase data bit No.20/19/18 position
W/R	4 / 0x6055	[14:0]	SUB_DATAPHASE0_17 _16_15	0x460f	Sub LCD 1 st phase data bit No.17/16/15 position
W/R	4 / 0x6056	[14:0]	SUB_DATAPHASE0_14 _13_12	0x39ac	Sub LCD 1 st phase data bit No.14/13/12 position
W/R	4 / 0x6057	[14:0]	SUB_DATAPHASE0_11 _10_9	0x2d49	Sub LCD 1 st phase data bit No.11/10/9 position
W/R	4 / 0x6058	[14:0]	SUB_DATAPHASE0_8_ 7_6	0x20e6	Sub LCD 1 st phase data bit No.8/7/6 position

W/R	4 / 0x6059	[14:0]	SUB_DATAPHASE0_5_ 4_3	0x1483	Sub LCD 1 st phase data bit No.5/4/3 position
W/R	4 / 0x605A	[14:0]	SUB_DATAPHASE0_2_ 1_0	0x0820	Sub LCD 1 st phase data bit No.2/1/0 position
W/R	4 / 0x605B	[14:0]	SUB_DATAPHASE1_23_ _22_21	0x5ed5	Sub LCD 2 nd phase data bit No.23/22/21 position
W/R	4 / 0x605C	[14:0]	SUB_DATAPHASE1_20_ _19_18	0x5272	Sub LCD 2 nd phase data bit No.20/19/18 position
W/R	4 / 0x605D	[14:0]	SUB_DATAPHASE1_17_ _16_15	0x460f	Sub LCD 2 nd phase data bit No.17/16/15 position
W/R	4 / 0x605E	[14:0]	SUB_DATAPHASE1_14_ _13_12	0x39ac	Sub LCD 2 nd phase data bit No.14/13/12 position
W/R	4 / 0x605F	[14:0]	SUB_DATAPHASE1_11_ _10_9	0x2d49	Sub LCD 2 nd phase data bit No.11/10/9 position
W/R	4 / 0x6060	[14:0]	SUB_DATAPHASE1_8_ 7_6	0x20e6	Sub LCD 2 nd phase data bit No.8/7/6 position
W/R	4 / 0x6061	[14:0]	SUB_DATAPHASE1_5_ 4_3	0x1483	Sub LCD 2 nd phase data bit No.5/4/3 position
W/R	4 / 0x6062	[14:0]	SUB_DATAPHASE1_2_ 1_0	0x0820	Sub LCD 2 nd phase data bit No.2/1/0 position
W/R	4 / 0x6063	[14:0]	SUB_DATAPHASE2_23_ _22_21	0x5ed5	Sub LCD 3 rd phase data bit No.23/22/21 position
W/R	4 / 0x6064	[14:0]	SUB_DATAPHASE2_20_ _19_18	0x5272	Sub LCD 3 rd phase data bit No.20/19/18 position
W/R	4 / 0x6065	[14:0]	SUB_DATAPHASE2_17_ _16_15	0x460f	Sub LCD 3 rd phase data bit No.17/16/15 position
W/R	4 / 0x6066	[14:0]	SUB_DATAPHASE2_14_ _13_12	0x39ac	Sub LCD 3 rd phase data bit No.14/13/12 position
W/R	4 / 0x6067	[14:0]	SUB_DATAPHASE2_11_ _10_9	0x2d49	Sub LCD 3 rd phase data bit No.11/10/9 position
W/R	4 / 0x6068	[14:0]	SUB_DATAPHASE2_8_ 7_6	0x20e6	Sub LCD 3 rd phase data bit No.8/7/6 position
W/R	4 / 0x6069	[14:0]	SUB_DATAPHASE2_5_ 4_3	0x1483	Sub LCD 3 rd phase data bit No.5/4/3 position
W/R	4 / 0x606A	[14:0]	SUB_DATAPHASE2_2_ 1_0	0x0820	Sub LCD 3 rd phase data bit No.2/1/0 position

W/R	4 / 0x606B	[3:0]	LCD_DITHER_CTRL	0x8	LCD dither control [3:1] : Display color resolution (with dither on) 7 : Reserved 6 : Reserved 5 : 24 bit (RGB888) 4 : 18 bit (RGB666) 3 : 16 bit (RGB565) 2 : 12 bit (RGB444) 1 : 8 bit (RGB332) 0 : 6 bit (RGB222) [0] : Dither on/off 1 : Dither on 0 : Dither off
W/R	4 / 0x606C	[0]	TV_RST	0	TV reset This bit is used to reset the TV encoder using external pin control. Writing “1” to this register makes TV_RST pin goes to “1”. Writing “0” to this register makes TV_RST pin goes to “0”.
W/R	4 / 0x606D	[2:0]	TV_AUTOCMD_NUM	0x0	TV auto command number
W/R	4 / 0x606E	[4:0]	TV_ADS	0x00	TV auto command ADS level
W/R	4 / 0x606F	[15:0]	TV_ADDR_DATA0	0x0000	TV 1 st auto command addr/data
W/R	4 / 0x6070	[15:0]	TV_ADDR_DATA1	0x0000	TV 2 nd auto command addr/data
W/R	4 / 0x6071	[12:0]	TV_IMGABORT_CNT	0x000	TV IMGABORT command wait count
W/R	4 / 0x6072	[15:0]	TV_ADDR_DATA2_1	0x0000	TV 3 rd auto command 1 addr/data
W/R	4 / 0x6073	[15:0]	TV_ADDR_DATA2_2	0x0000	TV 3 rd auto command 2 addr/data
W/R	4 / 0x6074	[15:0]	TV_ADDR_DATA3	0x0000	TV 4 th auto command addr/data
W/R	4 / 0x6075	[15:0]	TV_ADDR_DATA4_1	0x0000	TV 5 th auto command 1 addr/data

W/R	4 / 0x6076	[15:0]	TV_ADDR_DATA4_2	0x0000	TV 5 th auto command 2 addr/data
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OSD/Effect Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x7000	[1:0]	BG_OSD_CTRL	0x0	Background OSD control [1] : Background OSD mode 0 : Super impose 1 : Alpha blending [0] : Background OSD enable 0 : Background OSD disable 1 : Background OSD enable
W/R	4 / 0x7001	[11:0]	BG_OSD_H_SIZE	0x000	Background OSD horizontal size
W/R	4 / 0x7002	[11:0]	BG_OSD_V_SIZE	0x000	Background OSD vertical size
W/R	4 / 0x7003	[11:0]	BG_OSD_H_OFFSET	0x000	Background OSD horizontal offset
W/R	4 / 0x7004	[11:0]	BG_OSD_V_OFFSET	0x000	Background OSD vertical offset
W/R	4 / 0x7005	[9:0]	BG_OSD_START_ADD RH	0x00	Background OSD start address high [25:16]
W/R	4 / 0x7006	[15:0]	BG_OSD_START_ADD RL	0x0000	Background OSD start address low [15:0]
W/R	4 / 0x7007	[4:0]	BG_OSD_ALPHA	0x00	Background OSD alpha value
W/R	4 / 0x7008	[15:0]	BG_OSD_PATTERN	0x0000	Background OSD pattern value
W/R	4 / 0x7009	[2:0]	MN_OSD0_CTRL	0x0	Menu OSD 0 control [2:1] : Menu OSD 0 mode 0 : Super impose 1 : Alpha blending 2 : Super impose + Alpha blending [0] : Menu OSD 0 enable 0 : Menu OSD 0 disable 1 : Menu OSD 0 enable

W/R	4 / 0x700A	[11:0]	MN_OSD0_H_SIZE	0x000	Menu OSD 0 horizontal size
W/R	4 / 0x700B	[11:0]	MN_OSD0_V_SIZE	0x000	Menu OSD 0 vertical size
W/R	4 / 0x700C	[11:0]	MN_OSD0_H_OFFSET	0x000	Menu OSD 0 horizontal offset
W/R	4 / 0x700D	[11:0]	MN_OSD0_V_OFFSET	0x000	Menu OSD 0 vertical offset
W/R	4 / 0x700E	[9:0]	MN_OSD0_START_AD DRH	0x00	Menu OSD 0 start address high [25:16]
W/R	4 / 0x700F	[15:0]	MN_OSD0_START_AD DRL	0x0000	Menu OSD 0 start address low [15:0]
W/R	4 / 0x7010	[4:0]	MN_OSD0_ALPHA	0x00	Menu OSD 0 alpha value
W/R	4 / 0x7011	[15:0]	MN_OSD0_PATTERN0	0x0000	Menu OSD 0 pattern value 0
W/R	4 / 0x7012	[2:0]	MN_OSD1_CTRL	0x0	Menu OSD 1 control [2:1] : Menu OSD 1 mode 0 : Super impose 1 : Alpha blending 2 : Super impose + Alpha blending [0] : Menu OSD 1 enable 0 : Menu OSD 1 disable 1 : Menu OSD 1 enable
W/R	4 / 0x7013	[11:0]	MN_OSD1_H_SIZE	0x000	Menu OSD 1 horizontal size
W/R	4 / 0x7014	[11:0]	MN_OSD1_V_SIZE	0x000	Menu OSD 1 vertical size
W/R	4 / 0x7015	[11:0]	MN_OSD1_H_OFFSET	0x000	Menu OSD 1 horizontal offset
W/R	4 / 0x7016	[11:0]	MN_OSD1_V_OFFSET	0x000	Menu OSD 1 vertical offset
W/R	4 / 0x7017	[9:0]	MN_OSD1_START_AD DRH	0x00	Menu OSD 1 start address high [25:16]
W/R	4 / 0x7018	[15:0]	MN_OSD1_START_AD DRL	0x0000	Menu OSD 1 start address low [15:0]
W/R	4 / 0x7019	[4:0]	MN_OSD1_ALPHA	0x00	Menu OSD 1 alpha value

W/R	4 / 0x701A	[15:0]	MN_OSD1_PATTERN0	0x0000	Menu OSD 1 pattern value 0
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W/R	4 / 0x701B	[2:0]	MN_OSD2_CTRL	0x0	Menu OSD 2 control [2:1] : Menu OSD 2 mode 0 : Super impose 1 : Alpha blending 2 : Super impose + Alpha blending [0] : Menu OSD 2 enable 0 : Menu OSD 2 disable 1 : Menu OSD 2 enable
W/R	4 / 0x701C	[11:0]	MN_OSD2_H_SIZE	0x000	Menu OSD 2 horizontal size
W/R	4 / 0x701D	[11:0]	MN_OSD2_V_SIZE	0x000	Menu OSD 2 vertical size
W/R	4 / 0x701E	[11:0]	MN_OSD2_H_OFFSET	0x000	Menu OSD 2 horizontal offset
W/R	4 / 0x701F	[11:0]	MN_OSD2_V_OFFSET	0x000	Menu OSD 2 vertical offset
W/R	4 / 0x7020	[9:0]	MN_OSD2_START_AD DRH	0x00	Menu OSD 2 start address high [25:16]
W/R	4 / 0x7021	[15:0]	MN_OSD2_START_AD DRL	0x0000	Menu OSD 2 start address low [15:0]
W/R	4 / 0x7022	[4:0]	MN_OSD2_ALPHA	0x00	Menu OSD 2 alpha value
W/R	4 / 0x7023	[15:0]	MN_OSD2_PATTERN0	0x0000	Menu OSD 2 pattern value 0
W/R	4 / 0x7024	[2:0]	MN_OSD3_CTRL	0x0	Menu OSD 3 control [2:1] : Menu OSD 3 mode 0 : Super impose 1 : Alpha blending 2 : Super impose + Alpha blending [0] : Menu OSD 3 enable 0 : Menu OSD 3 disable 1 : Menu OSD 3 enable
W/R	4 / 0x7025	[11:0]	MN_OSD3_H_SIZE	0x000	Menu OSD 3 horizontal size

W/R	4 / 0x7026	[11:0]	MN_OSD3_V_SIZE	0x000	Menu OSD 3 vertical size
W/R	4 / 0x7027	[11:0]	MN_OSD3_H_OFFSET	0x000	Menu OSD 3 horizontal offset
W/R	4 / 0x7028	[11:0]	MN_OSD3_V_OFFSET	0x000	Menu OSD 3 vertical offset
W/R	4 / 0x7029	[9:0]	MN_OSD3_START_AD DRH	0x00	Menu OSD 3 start address high [25:16]
W/R	4 / 0x702A	[15:0]	MN_OSD3_START_AD DRL	0x0000	Menu OSD 3 start address low [15:0]
W/R	4 / 0x702B	[4:0]	MN_OSD3_ALPHA	0x00	Menu OSD 3 alpha value
W/R	4 / 0x702C	[15:0]	MN_OSD3_PATTERN0	0x0000	Menu OSD 3 pattern value 0
W/R	4 / 0x702D	[3:0]	MIRROR_FLIP_ROTAT E	0x00	Mirror, flip and rotate [3] : Mirror 0 : Nor mirror 1 : Mirror [2] : Flip 0 : Not flip 1 : Flip [1:0] : Rotate 3 : Rotate 270° 2 : Rotate 180° 1 : Rotate 90° 0 : Not rotate

W/R	4 / 0x7030	[10:0]	EFFECT_CTRL	0x00	Effect control [10] : Contrast enable 0 : Contrast disable 1 : Contrast enable [9] : Hue enable 0 : Hue disable 1 : Hue enable [8] : Brightness enable 0 : Brightness disable 1 : Brightness enable [7] : Saturation enable 0 : Saturation disable 1 : Saturation enable [6] : Emboss effect enable 0 : Emboss effect disable 1 : Emboss effect enable [5] : Sketch effect enable 0 : Sketch effect disable 1 : Sketch effect enable [4] : Lightning effect enable 0 : Lightning effect disable 1 : Lightning effect enable [3] : Sepia effect enable 0 : Sepia effect disable 1 : Sepia effect enable [2] : Fade effect enable 0 : Fade effect disable 1 : Fade effect enable [1] : Solarization effect enable 0 : Solarization effect disable 1 : Solarization effect enable [0] : Negative effect enable 0 : Negative effect disable 1 : Negative effect enable
W/R	4 / 0x7031	[0]	SOL_CTRL	0	Solarization control [0] : Solarization color invert
W/R	4 / 0x7032	[7:0]	SOL_MASK_LEVEL	0x1f	Solarization mask level

W/R	4 / 0x7033	[7:0]	SOL_MASK_GAIN	0x06	Solarization mask gain
W/R	4 / 0x7034	[7:0]	FADE_LEVEL	0x50	Fade level
W/R	4 / 0x7035	[1:0]	SEP_RATE	0x0	Sepia rate
W/R	4 / 0x7036	[7:0]	SEP_CB_VALUE	0x2c	Sepia Cb value
W/R	4 / 0x7037	[7:0]	SEP_CR_VALUE	0xbe	Sepia Cr value
W/R	4 / 0x7038	[7:0]	LIGHT_EXT_MUL_GAIN	0x40	Lightning external multi gain
W/R	4 / 0x7039	[7:0]	LIGHT_CEN_MUL_GAIN	0xc0	Lightning center multi gain
W/R	4 / 0x703A	[9:0]	LIGHT_X_AXIS	0x100	Lightning X axis
W/R	4 / 0x703B	[9:0]	LIGHT_Y_AXIS	0x0f0	Lightning Y axis
W/R	4 / 0x703C	[9:0]	LIGHT_RADIUS	0x050	Lightning radius
W/R	4 / 0x703D	[7:0]	SKETCH_CTRL	0x28	Sketch control [7] : Sketch color on [6:5] : Sketch pencil width [4:0] : Sketch core value
W/R	4 / 0x703E	[0]	EMBOSS_CTRL	0	Emboss control [0] : Emboss color on
W/R	4 / 0x7040	[15:0]	MN_OSD0_PATTERN1	0x0000	Menu OSD 0 pattern value 1
W/R	4 / 0x7041	[15:0]	MN_OSD1_PATTERN1	0x0000	Menu OSD 1 pattern value 1
W/R	4 / 0x7042	[15:0]	MN_OSD2_PATTERN1	0x0000	Menu OSD 2 pattern value 1
W/R	4 / 0x7043	[15:0]	MN_OSD3_PATTERN1	0x0000	Menu OSD 3 pattern value 1

W/R	4 / 0x7044	[1:0]	PIP_CTRL	0x0	PIP control [1] : PIP mode 0 : Super impose 1 : Alpha blending [0] : PIP enable 0 : PIP disable 1 : PIP enable
W/R	4 / 0x7045	[11:0]	PIP_H_SIZE	0x000	PIP horizontal size
W/R	4 / 0x7046	[11:0]	PIP_V_SIZE	0x000	PIP vertical size
W/R	4 / 0x7047	[11:0]	PIP_H_OFFSET	0x000	PIP horizontal offset
W/R	4 / 0x7048	[11:0]	PIP_V_OFFSET	0x000	PIP vertical offset
W/R	4 / 0x7049	[9:0]	PIP_START_ADDRH	0x00	PIP start address high [25:16]
W/R	4 / 0x704A	[15:0]	PIP_START_ADDRL	0x0000	PIP start address low [15:0]
W/R	4 / 0x704B	[4:0]	PIP_ALPHA	0x00	PIP alpha value
W/R	4 / 0x704C	[15:0]	PIP_COLOR_TABLE0	0x0000	PIP color table 0
W/R	4 / 0x704D	[15:0]	PIP_COLOR_TABLE1	0x0000	PIP color table 1
W/R	4 / 0x704E	[15:0]	PIP_COLOR_TABLE2	0x0000	PIP color table 2
W/R	4 / 0x704F	[15:0]	PIP_COLOR_TABLE3	0x0000	PIP color table 3
W/R	4 / 0x7050	[15:0]	PIP_COLOR_TABLE4	0x0000	PIP color table 4
W/R	4 / 0x7051	[15:0]	PIP_COLOR_TABLE5	0x0000	PIP color table 5
W/R	4 / 0x7052	[15:0]	PIP_COLOR_TABLE6	0x0000	PIP color table 6
W/R	4 / 0x7053	[15:0]	PIP_COLOR_TABLE7	0x0000	PIP color table 7

W/R	4 / 0x7054	[7:0]	EDGE_LOWLEVEL	0x00	Edge Low level control
W/R	4 / 0x7055	[3:0]	EDGE_MULGAIN	0x0	Edge multiply gain
W/R	4 / 0x7056	[3:0]	EDGE_EGAIN	0x7	Edge Level shift
W/R	4 / 0x7057	[3:0]	EDGE_LCORE	0x8	Edge Gain ratio
W/R	4 / 0x7058	[7:0]	EDGE_CTRL	0x68	Edge control

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Memory Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x8000	[2:0]	DIR_MEM_ACC_MOD E	0x0	Direct memory access mode [2] : Data bit 0 : 16 bit 1 : 32 bit [1] : Reserved [0] : Write/Read mode 0 : Write mode 1 : Read mode
W/R	4 / 0x8001	[9:0]	DIR_MEM_START_AD DRH	0x00	Direct memory start address high [25:16]
W/R	4 / 0x8002	[15:0]	DIR_MEM_START_AD DRL	0x0000	Direct memory start address low [15:0]
W/R	4 / 0x8003	[15:0]	DIR_MEM_DATA	0x0000	Direct memory data
W/R	4 / 0x8004	[15:0]	DIR_MEM_DATA_SIZE H	0x0000	Direct memory data size high [31:16]
W/R	4 / 0x8005	[15:0]	DIR_MEM_DATA_SIZE L	0x0000	Direct memory data size low [15:0]

System Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0x9000	[0]	REG_OWN_SEL	(1 for parallel) / (0 for SPI to UART)	Register ownership select 0: Internal MCU has the ownership of register access 1: External HOST has the ownership of register access
W/R	4 / 0x9001	[14:0]	DELAY_CTRL	0x0000	System delay control [14:12] : TV CS delay 7 : 7 delay, 6 : 6 delay, 5 : 5 delay, 4 : 4 delay, 3 : 3 delay, 2 : 2 delay, 1 : 1 delay, 0 : 0 delay [11:9] : Graphic addr0 delay 7 : 7 delay, 6 : 6 delay, 5 : 5 delay, 4 : 4 delay, 3 : 3 delay, 2 : 2 delay, 1 : 1 delay, 0 : 0 delay [8:6] : RS delay 7 : 7 delay, 6 : 6 delay, 5 : 5 delay, 4 : 4 delay, 3 : 3 delay, 2 : 2 delay, 1 : 1 delay, 0 : 0 delay [5:3] : Main LCD CS delay 7 : 7 delay, 6 : 6 delay, 5 : 5 delay, 4 : 4 delay, 3 : 3 delay, 2 : 2 delay, 1 : 1 delay, 0 : 0 delay [2:0] : Sub LCD CS delay 7 : 7 delay, 6 : 6 delay, 5 : 5 delay, 4 : 4 delay, 3 : 3 delay, 2 : 2 delay, 1 : 1 delay, 0 : 0 delay
W/R	4 / 0x9002	[10:0]	PLL_M	??	PLL main divider parameter

W/R	4 / 0x9003	[7:0]	PLL_P	??	PLL pre divider parameter
W/R	4 / 0x9005	[0]	SYS_CTRL	0	System control [0] : Data Endian 0 : Little endian for parallel host I/F data 1 : Big endian for parallel host I/F data
W/R	4 / 0x9006	[0]	STARTUP_CODE_ DOWN_EN	0	Start-up code download enable Enable bit to write firmware code to internal code-RAM. Actual firmware data should be written to next STARTUP_CODE register after enable this register. 0: Parallel firmware download disable 1: Parallel firmware download enable
W/R	4 / 0x9007	[7:0]	STARTUP_CODE	0x00	Start-up code data External host writes firmware for internal MCU through this register. The firmware data written in this register will be written to internal 8KB code-RAM automatically. The RAM address is auto incremented during continuous register access.
W/R	4 / 0x9008	[7:0]	GPIO_OWN_SEL	0xFF	GPIO ownership select 0 : GPIO pins are controlled by internal MCU base on port access 1 : GPIO pins are controlled by external host base on register access
W/R	4 / 0x9009	[7:0]	GPIO_DIR	0xFF	GPIO direction 0 : Each GPIO act as output 1 : Each GPIO act as input
W/R	4 / 0x900A	[7:0]	GPIO_DATA	0x00	GPIO data Data value to be written to GPIO port when write mode (GPIO_DIR = '0') Data value comes from GPIO port when read mode (GPIO_DIR = '1')
W/R	4 / 0x900B	[1:0]	INT_MODE_CTRL	0x00	Interrupt mode control [1] : Interrupt type 0 : Edge trigger interrupt 1 : Level trigger interrupt [0] : Interrupt polarity

					0 : Active Low 1 : Active High
W/R	4 / 0x900C	[7:0]	INT_EN	0x00	Interrupt enable [15] : Host to 8051 command buffer error interrupt enable [7] : Graphic interrupt enable [6] : VSYNC interrupt enable [5] : Preview stop interrupt enable [4] : Decode error interrupt enable [3] : Decode interrupt enable [2] : Encode interrupt enable [1] : LCD frame update interrupt enable [0] : 8051 to host command buffer interrupt Enable
R	4 / 0x900D	[7:0]	INT_STATUS	0x00	Interrupt Status [15] : Host to 8051 command error [7] : Graphic operation complete [6] : VSYNC negative edge detect [5] : Preview stop complete [4] : Decoder error [3] : Decode complete [2] : Encode complete [1] : LCD frame update complete [0] : 8051 to host command complete
W	4 / 0x900E	[7:0]	INT_ACK	0x00	Interrupt acknowledge (Auto Cleared) [15] : Host to 8051 command error [7] : Graphic operation complete [6] : VSYNC negative edge detect [5] : Preview stop complete [4] : Decoder error [3] : Decode complete [2] : Encode complete [1] : LCD frame update complete [0] : 8051 to host command complete
	4 / 0x900F	[7:0]	GPIO_MODE	0x00	GPIO direction 0 : Each GPIO act as normal gpio 1 : Each GPIO act as special signal
W/R	4 / 0x9010	[1:0]	SOFT_RESET	0x3	Software reset [1] : All reset

					[0] : Core reset
W/R	4 / 0x9011	[5:0]	SUB_PWR_CTRL	0x00	Sub power control [5] : Upscale related block power down [4] : MCU related block power down [3] : Sensor interface related block power down [2] : JPEG encoder related block power down [1] : JPEG decoder related block power down [0] : DMA related block power down
W/R	4 / 0x9012	[15:0]	SYS_CLK_CTRL	0x0000	System clock control [15:12] : LCDCLK select [15] : LCDCLK select 0 : External clock 1 : PLL output clock [14:12] : LCDCLK divide 3 : PLL output clock / 8 2 : PLL output clock / 7 1 : PLL output clock / 6 0 : PLL output clock / 5 3 : PLL output clock / 4 2 : PLL output clock / 3 1 : PLL output clock / 2 0 : PLL output clock bypass [11:8] : MEMCLK select [11] : MEMCLK select 0 : External clock 1 : PLL output clock [10:8] : MEMCLK divide 3 : PLL output clock / 8 2 : PLL output clock / 7 1 : PLL output clock / 6 0 : PLL output clock / 5 3 : PLL output clock / 4 2 : PLL output clock / 3 1 : PLL output clock / 2 0 : PLL output clock bypass [7:4] : GRPCLK select [7] : GRPCLK select 0 : External clock

					1 : PLL output clock [6:4] : GRPCLK divide 3 : PLL output clock / 8 2 : PLL output clock / 7 1 : PLL output clock / 6 0 : PLL output clock / 5 3 : PLL output clock / 4 2 : PLL output clock / 3 1 : PLL output clock / 2 0 : PLL output clock bypass [3:0] : CAMCLK select [3] : CAMCLK select 0 : External clock 1 : PLL output clock [2:0] : CAMCLK divide 3 : PLL output clock / 8 2 : PLL output clock / 7 1 : PLL output clock / 6 0 : PLL output clock / 5 3 : PLL output clock / 4 2 : PLL output clock / 3 1 : PLL output clock / 2 0 : PLL output clock bypass
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Misc Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
R	4 / 0xA000	[15:0]	PRODUCT_ID	0x0011	Product ID number
R	4 / 0xA001	[15:0]	PRODUCT_REV	0x0001	Product revision number
W/R	4 / 0xA002	[15:0]	TEST_REG	0x5aa5	Host I/F write/read test register
W/R	4 / 0xA003	[0]	TEST_PATN_EN	0	Test pattern enable

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Command Control

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0xB000	[2:0]	CMD_CTRL	0x00	Command control [2] : Command FIFO clear 0 : Not clear internal pointer for CMD_DATA FIFO to prepare CMD_DATA write 1 : Clear internal pointer for CMD_DATA FIFO to prepare CMD_DATA write [1:0] : Command number Specify command length (number of the byte). The command sequence is started automatically after finishing write of this number of bytes to CMD_DATA register. 3 : CMD_DATA receives 4 bytes command data, then internal MCU execute the command sequence 2 : CMD_DATA receives 3 bytes command data, then internal MCU execute command the command sequence 1 : CMD_DATA receives 2 bytes command data, then internal MCU execute the command sequence 0 : CMD_DATA receives 1 byte command data, then internal MCU execute he command
W/R	4 / 0xB001	[7:0]	CMD_DATA	0x00	Command data
R	4 / 0xB101	[7:0]	CMD_RSTATUS	0x00	Command Read status
R	4 / 0xB102	[7:0]	CMD_RDATA	0x00	Command Read data
R	4 / 0xB103	[7:0]	CMD_RCMDNUMBER	0x00	A mount of Command

Point Processing

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	4 / 0xC000	[15:0]	LSC_CTRL		Lens shade compensation control [15:8] : Lens_mgain [7:4] : Reserved [3:1] : Lens_div [0] : Lens shade enable
W/R	4 / 0xC001	[9:0]	LSC_X_AXIS		
W/R	4 / 0xC002	[9:0]	LSC_Y_AXIS		
W/R	4 / 0xC003	[7:0]	SAT_BG	0x00	Brightness Gain [7] : Sign -- 0: negative , 1 : positive [6:0] : Magnitude
W/R	4 / 0xC004	[7:0]	SAT_SGB	0x80	Saturation BGain Gain value : 0 ~ 1.99
W/R	4 / 0xC005	[7:0]	SAT_SQA	0x80	Saturation RGain Gain value : 0 ~ 1.99
W/R	4 / 0xC006	[7:0]	HUE_CG	0x40	Contrast Gain Gain value : 0 ~ 3.99
W/R	4 / 0xC007	[7:0]	HUE_H22	0x40	Hue Gain (cb cos Ø) [7] : Sign -- 0: negative , 1 : positive [6:0] : Magnitude
W/R	4 / 0xC008	[7:0]	HUE_H23	0x00	Hue Gain (cr sin Ø) [7] : Sign -- 0: negative , 1 : positive [6:0] : Magnitude
W/R	4 / 0xC009	[7:0]	HUE_H32	0x00	Hue Gain(-cb sin Ø) [7] : Sign -- 0: negative , 1 : positive [6:0] : Magnitude
W/R	4 / 0xC00A	[7:0]	HUE_H33	0x40	Hue Gain(cr cos Ø) [7] : Sign -- 0: negative , 1 : positive [6:0] : Magnitude
W/R	4 / 0xC020	[11:0]	SENSOR_H_SIZE	0x0000	Sensor horizontal size
W/R	4 / 0xC021	[10:0]	SENSOR_V_SIZE	0x0000	Sensor vertical size

W/R	4 / 0xC022	[0]	AF_CTRL	0	AF control [0] : AF mode enable
W/R	4 / 0xC023	[6:0]	AF_AE_MODE_REG	0x00	AF/AE mode selection
W/R	4 / 0xC024	[2:0]	AF_LPF_MODE	0x0	AF LPF mode selection
W/R	4 / 0xC025	[7:0]	AF_CORE	0x00	AF Coring Value
W/R	4 / 0xC026	[4:0]	H33_COEF	0x1F	AF HPF33 tab variable coefficient
W/R	4 / 0xC027	[3:0]	H17_COEF	0xF	AF HPF17 tab variable coefficient
W/R	4 / 0xC028	[2:0]	H9_COEF	0x7	AF HPF9 tab variable coefficient
W/R	4 / 0xC029	[2:0]	H7_COEF	0x5	AF HPF7 tab variable coefficient
W/R	4 / 0xC02A	[0]	AE_LPF_MODE	1	AE LPF mode selection
W/R	4 / 0xC02B	[5:0]	AE_CLIP	0x00	AE clip value
W/R	4 / 0xC02C	[7:0]	AE_YH	0xF0	AE high threshold value
W/R	4 / 0xC02D	[7:0]	AE_YL	0x10	AE low threshold value
W/R	4 / 0xC02E	[7:0]	WINAF_VCE_A	0x00	Vertical center position AF windows A
W/R	4 / 0xC02F	[7:0]	WINAF_VSC_A	0x00	Vertical scale AF windows A
W/R	4 / 0xC030	[7:0]	WINAF_HCE_A	0x00	Horizontal center position AF windows A
W/R	4 / 0xC031	[7:0]	WINAF_HSC_A	0x00	Horizontal scale AF windows A
W/R	4 / 0xC032	[7:0]	WINAF_VCE_B	0x00	Vertical center position AF windows B
W/R	4 / 0xC033	[7:0]	WINAF_VSC_B	0x00	Vertical scale AF windows B
W/R	4 / 0xC034	[7:0]	WINAF_HCE_B	0x00	Horizontal center position AF windows B

W/R	4 / 0xC035	[7:0]	WINAF_HSC_B	0x00	Horizontal scale AF windows B
W/R	4 / 0xC036	[7:0]	WINAE_HST	0x00	Horizontal area start position AE windows
W/R	4 / 0xC037	[7:0]	WINAE_HED	0x00	Horizontal area end position AE windows
W/R	4 / 0xC038	[7:0]	WINAE_VST	0x00	Vertical area start position AE windows
W/R	4 / 0xC039	[7:0]	WINAE_VED	0x00	Vertical area end position AE windows
R	4 / 0xC03A	[15:0]	AF1SA_REG_H	0x00	AF1 Window A accumulate value High
R	4 / 0xC03B	[15:0]	AF1SA_REG_L	0x00	AF1 Window A accumulate value Low
R	4 / 0xC03C	[15:0]	AF1SB_REG_H	0x00	AF1 Window A accumulate value High
R	4 / 0xC03D	[15:0]	AF1SB_REG_L	0x00	AF1 Window A accumulate value Low
R	4 / 0xC03E	[15:0]	AF2SA_REG_H	0x00	AF2 Window A accumulate value High
R	4 / 0xC03F	[15:0]	AF2SA_REG_L	0x00	AF2 Window A accumulate value Low
R	4 / 0xC040	[15:0]	AF2SB_REG_H	0x00	AF2 Window A accumulate value High
R	4 / 0xC041	[15:0]	AF2SB_REG_L	0x00	AF2 Window A accumulate value Low
R	4 / 0xC042	[15:0]	AFEDGE_A_REG_H	0x00	AF Edge Window A accumulate value High
R	4 / 0xC043	[15:0]	AFEDGE_A_REG_L	0x00	AF Edge Window A accumulate value Low
R	4 / 0xC044	[15:0]	AFEDGE_B_REG_H	0x00	AF Edge Window B accumulate value High
R	4 / 0xC045	[15:0]	AFEDGE_B_REG_L	0x00	AF Edge Window B accumulate value Low
R	4 / 0xC046	[15:0]	AE_ACC_REG_H	0x00	AE Window accumulate value High
R	4 / 0xC047	[15:0]	AE_ACC_REG_L	0x00	AE Window accumulate value Low

R	4 / 0xC048	[15:0]	AE_CNT_REG_H	0x00	AE Window data Count High
R	4 / 0xC049	[15:0]	AE_CNT_REG_L	0x00	AE Window data Count Low

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BANK5 (M_ADDR[4:1] == 0x05)**Graphic Register & SDRAM controller Control**

Access	Bank/ Sub- Address	Bit	Name	Default	Description
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BANK6 (M_ADDR[4:1] == 0x06)

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	2 / Direct	[15:0]	TV_DATA		Used when TV control bypass mode. The D[15:0], WR, RD, CS signals are bypassed to LD[15:0], L_WR, L_RD, L_CS_S with this address..

BANK7 (M_ADDR[4:1] == 0x07)**Power Control**

Access	Bank/ Sub- Address	Bit	Name	Default	Description
W/R	7 / Direct	[7:0]	MAIN_POWER_CON	0x14	[7] SENSOR_PWDN_R Right sensor power down [6] CAMERA_ALL_PWDN All camera parts power down except MCU, PLL, External SDRAM and SENSOR [5] LCDIF_PWDN LCD interface power down [4] TV_PWDN NTSC/PAL encoder power down [3] SDRAMC_PWDN SDRAM controller power down [2] APLL_PWDN On-chip Analog PLL power down [1] OSC_CTRL_EN Oscillator control enable [0] SENSOR_PWDN_L Left sensor power down.